

Degree Project in Materials Science and Engineering

Second Cycle (Master of Science), 30 Credits

Comparative Analysis of Accelerated Corrosion Chambers in Automotive Engineering

An Evaluation of the Influence of Design Parameters on Test Homogeneity and Outcomes at Volvo Cars

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Abstract

Accelerated corrosion testing is a cornerstone of automotive component qualification, enabling the prediction of years of in-service degradation within compressed laboratory timeframes. At Volvo Cars, the ACT II (Accelerated Corrosion Test II) (VCS 1027,1449) is the primary laboratory test for this purpose, nominally replicating six years of field exposure within a six to twelve-week test. A fundamental but underexamined assumption in this framework is whether the results obtained from testing samples using one chamber are directly comparable to those obtained using another chamber produced by a different supplier, and whether cross-laboratory reproducibility holds.

This master thesis investigates whether two commercial accelerated corrosion chambers, the ControlArt TC210 and the Ascott Atmosfär Premium AT1300ip, deliver equivalent ACT II test outcomes, and develops a structured, sample-based methodology to assess chamber comparability and qualification readiness. The work is further motivated by a temperature malfunction that occurred in the Ascott chamber during the test campaign, preventing it from consistently reaching multiple temperature setpoints. Rather than a disqualifying failure, this deviation was treated as a variable for evaluating the sensitivity of ACT II outcomes to setpoint non-compliance and the adequacy of the standard's tolerance bands.

The study was structured into five work packages (WP1–WP5) of increasing complexity: calibration (WP1), zinc foil chessboard indicator testing for spatial homogeneity mapping (WP2), scribed painted panel corrosion propagation across four material systems (WP3), crevice panel testing using galvanized steel (GI 50/50) coupled with boron steel (WP4), and evaluation of real automotive sub-assemblies including brake hoses (WP5). Evaluation methods include gravimetric analysis, optical microscopy, CT scanning, and statistical analysis. Combined, these techniques provided both macroscopic and microscopic characterization of corrosion behavior.

Generated results demonstrate measurable differences between the two corrosion chambers (platforms) in temperature uniformity, humidity control, and spatial aggressiveness distribution, with the Ascott malfunction producing a quantifiable bias in certain work packages. The comparative methodology developed here provides a reusable chamber qualification workflow applicable to newly installed or repaired chambers and identifies specific aspects of the ACT II standard where calibration requirements lack sufficient specificity. The findings lay the groundwork for future extension to walk-in chambers and the possible ACT III cycle elaborated by Volvo Cars for large automotive components.

Keywords

Accelerated corrosion testing; ACT II; corrosion chamber; crevice corrosion; galvanic corrosion; automotive materials; Volvo Cars.

Sammanfattning

Accelererad korrosionsprovning är en hörnsten i kvalificering av fordonskomponenter eftersom den möjliggör förutsägelse av flera års nedbrytning i drift inom komprimerade laboratorietidsramar. Hos Volvo Cars är det cykliska ACT II-protokollet (VCS 1027,1449) det primära verktyget för detta ändamål och syftar nominellt till att återskapa sex års kloridexponering inom en testkampanj på sex till tolv veckor. Ett grundläggande, men otillräckligt undersökt, antagande i detta ramverk är att resultat som erhålls i en kammarplattform är direkt jämförbara med resultat från provning hos en annan leverantör, samt att reproducerbarhet mellan laboratorier kan uppträda.

Detta examensarbete undersöker huruvida två kommersiella accelererade korrosionskammare, ControlArt TC210 och Ascott Atmosfär Premium AT1300ip, ger likvärdiga ACT II-testresultat, samt utvecklar en strukturerad, provbaserad metodik för att bedöma kammarjämförbarhet och kvalificeringsberedskap. Arbetet motiveras ytterligare av ett temperaturfel som inträffade i Ascott-kammaren under testkampanjen vilket hindrade den från att konsekvent uppnå flera temperaturbörvärden. I stället för att betrakta denna avvikelse som ett diskvalificerande fel, behandlas den som en variabel för att utvärdera ACT II-resultatens känslighet för bristande uppfyllelse av börvärden samt lämpligheten hos standardens toleransintervall.

Studien var strukturerad i fem arbetspaket (WP1–WP5) med ökande komplexitet: miljökarakterisering av bägge kamrar, zinkfolieprovning med schackbrädesindikator för kartläggning av rumslig homogenitet, korrosionsutbredning från ritsade lackerade paneler över fyra materialsystem, spaltkorrosionsprovning med galvaniserat stål (GI 50/50) kopplat till borstål, samt utvärdering av verkliga fordonsdelkomponenter inklusive bromsslangar. Utvärderingsmetoderna omfattar gravimetrisk analys, optisk mikroskopi, CT-skanning och statistisk analys, metoder vilka tillsammans ger både makroskopisk och mikroskopisk karakterisering av korrosionsbeteendet.

Resultaten visar mätbara skillnader mellan de två plattformarna avseende temperaturjämnhet, fuktighetskontroll och rumslig fördelning av aggressivitet, där funktionsfelet i Ascott-kammaren gav upphov till en kvantifierbar systematisk avvikelse i vissa arbetspaket. Den jämförande metodik som utvecklats inom ramen för examensarbetet tillhandahåller ett återanvändbart arbetsflöde för kvalificering av kammare, tillämpbart på nyinstallerade eller reparerade kammare, samt identifierar specifika delar av ACT II-standarden där kalibreringskraven saknar tillräcklig tydlighet. Resultaten lägger grunden för framtida utvidgning till ”walk-in-kammare” och för ACT III-cykeln som tas fram vid Volvo Cars för stora fordonskomponenter.

Nyckelord

Accelererad korrosionsprovning; ACT II; korrosionskammare; spaltkorrosion; galvanisk korrosion; fordonsmaterial; Volvo Cars

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Table of contents

Comparative Analysis of Accelerated Corrosion Chambers in Automotive Engineering i
An Evaluation of the Influence of Design Parameters on Test Homogeneity and Outcomes at Volvo Cars i
Abstract iv
Keywords iv
Sammanfattning v
Nyckelord v
Acknowledgments vii
Table of contents vii
List of Figures xi
List of Tables xiii
List of acronyms and abbreviations xv
1 Introduction and Background 1
1.1 Problem 2
1.2 Purpose 2
1.3 Goals 3
1.4 Research Methodology 3
1.5 Delimitations 4
2 Background 5
2.1 Fundamentals of Corrosion 5
Electrochemical Nature of Corrosion 5
Thermodynamics of Corrosion 6
Kinetics of Corrosion 6
Passivation 6
2.2 Factors Influencing Corrosion Rates 6
Temperature 6
pH 7
Salinity and Chloride Ion Concentration 7
Metallurgical Factors 7
2.3 Principal Types of Corrosion 7
Uniform (General) Corrosion 7
Pitting Corrosion 8
Galvanic Corrosion 8
Crevice Corrosion 8
Stress Corrosion Cracking (SCC) 8
2.4 Accelerated Corrosion Test Chambers 8
Principles of Accelerated Testing 8
Fundamental Parameters Governing Corrosion Chamber Performance: Temperature control and humidity 9
Cyclic Test Programme Control 10
The ACT II (Accelerated Corrosion Test, Version II) Cycle 10
2.5 Comparison of Test Chambers: ControlArt TC210 and Ascott Atmosfär Premium AT1300ip 11

ControlArt TC210 11
Ascott Atmosfär Premium AT1300ip 11
Comparative Analysis 12
3 Methodologies and Methods 13
3.1 Research Process 13
3.2 Research Paradigm and Philosophical Position 13
3.3 Sample structure 13
Specimen Types and Work Package Structure 13
Specimen Preparation 14
Exposure and Weekly Inspection 14
3.4 Measurement Methods 14
Gravimetric Analysis 14
Cross-Section Optical Microscopy 14
CT Scanning 15
ATR-FTIR Spectroscopy 15
Scanning Electrochemical Microscopy and Localised Techniques 15
Statistical Analysis 16
3.5 Reliability and Validity 16
Reliability 16
Validity 16
3.6 Ethical and Societal Considerations 17
3.7 Research Process Scheme 17
4 Experimental Work: Specimen Description and Preparation 18
4.1 WP1: Calibration 18
4.2 WP2: Zinc tiles 18
4.3 Scribed panels 20
Panel preparation and scribing 21
4.4 WP4: Crevice Panels 22
4.5 WP5: Complex components 22
5 Results and Analysis 24
5.1 WP1: Calibration 24
Control Art TC210 29
Comparative Analysis: Ascott vs ControlArt 30
5.2 WP2: Zinc Foil 32
Statistical Summary 33
Chamber Comparison 33
Corner Homogeneity 34
Pit Depth Analysis 34
Per-Panel Statistics 35
Grouped Statistics and the Centre-to-Corner Gradient 36
Chamber Comparison and Mechanism Explanation 36
Key Takeaways and Chamber Assessment 37
5.3 WP3: Scribed Panels 38
Descriptive statistics 38
Subgroup Breakdown by Material, Chamber and Orientation 39
Scribe Orientation Effect 40
Variability Analysis: Coefficient of Variation 41
Chamber Comparison 43

Materials With No Chamber Effect 43
ZM2406F08....43
H2507F08 44
Summary and Conclusions....45
5.4 WP4: Crevice Panels....46
Overview of Post-Exposure Appearance....46
Phosphate (PH) specimens....47
CT Scan Assessment....47
5.5 WP5: Complex Components....48
6 ACT II suggested Improvement 48
Standardised Chamber Loading Condition During Calibration....49
7 Research questions answered 50
8 Use of generative AI 51
9 Future work 51
9.1 Future work....51
References.... 54

List of Figures

Figure 1: ACT II - Complete weekly cycles in temperature and RH 11
Figure 2: Research Process Scheme 17
Figure 3: Zinc tiles samples collection before testing 19
Figure 4: Zinc tiles located at the Ascott's chamber test area corners after a 6-week cycle testing, before deep cleaning. 19
Figure 5: CT scan image of Sample 1 after 2 weeks to the ACT II cycles 20
Figure 6: Zinc tiles positioning scheme in both test chambers 20
Figure 7: Scribed panels after 4 weeks of ACT II cycle in the ControlArt chamber 21
Figure 8: Example of corrosion of panels with crevices after 6 weeks of ACT2 in the xx chamber 22
Figure 9 Ascott brake hose after full test 23
Figure 10 Brake hose sample holder 23
Figure 11: Ascott chamber ACT II cycle log of the temperature and the relative humidity 24
Figure 12 Ascott ACT IIcycle after the first intervention 25
Figure 13: Temperature and relative humidity log from internal and external sensor in the Ascott chamber 26
Figure 14: Differences between the two external probes. With Sensor 1 being the central one. 31
Figure 15: Example of cross cut view of a corroded zinc tile exposed for 6 weeks to ACTII at the optical microscope 35
Figure 16 Test crevice panels exposed in the Ascottand ControlArt chambers for 6 weeks of the ACT II Cycle 46
Figure 17 Scan of a thin sample exposed to ACTII 47
Figure 18: Brake hose Cross cut comparison 48

List of Tables

Table 1: Parameter comparison: ControlArt TC210 vs Ascott Atmosfär Premium AT1300ip. 12
Table 2: Ascott AT1300ip — setpoint analysis results for all three sensors and three setpoints. DAS = mean offset from nominal; MSPDA = maximum short-term fluctuation. Limits: DAS ±0.4 °C / ±2% RH, MSPDA ±0.6 °C / ±3% RH. 26
Table 3: Ascott AT1300ip, time to cool from 50 °C to 25 °C. VCS 1027,1449 limit: 60 minutes. 27
Table 4: Ascott AT1300ip, absolute spatial difference |S2 − S1| over the full calibration run. 28
Table 5: ControlArt TC210 — setpoint analysis results for all three sensors and three setpoints. Same limits as Table 5.1. 29
Table 6: ControlArt TC210: absolute spatial difference |M01 − M00| over all sessions combined. 30
Table 7: Corrosion product (g) per tile, sorted by position group. 32
Table 8: Statistical summary of the measured mass of corrosion products listed by the position group and chamber for the zinc tiles. IQR = interquartile range (robust spread measure not affected by outlier tiles). 32
Table 9: Chamber comparison of the measured mass of corrosion product listed by the position group of exposed zinc tiles. ΔCV% = AS CV% − CA CV%; negative values (pp = percentage points) indicate ControlArt resulted in more variable findings for that group. 33
Table 10: Comparison of corner-by-corner mass of corrosion product formed after xx days exposure to the ACT II cycle and within-chamber corner homogeneity statistics. 34
Table 11 Pit depths and number on zinc tiles exposed. Per-panel pit depth statistics. n = individual pits measured. 35
Table 12: Chamber comparison for pit depth of the zinc tiles….. Positive CA−AS values indicate that exposure in the ControlArt chamber resulted in deeper or more variable pits. 36
Table 13: Complete summary of key metrics 37
Table 14: Overall descriptive statistics by material (n = 12 position averages per material, across both chambers and both orientations). * CV% for A2202F03 is not interpretable due to near-zero mean. 39
Table 15: Orientation effect of the scribed materials exposed….. : mean propagation for diagonal (D) and horizontal (H) scribes, with H−D difference. Positive H−D values indicate horizontal scribe was more severely corroded. 40
Table 16: Coefficient of variation (CV%) by material × chamber × orientation (n = 3 per subgroup). * CV% not interpretable for A2202F03 due to near-zero mean. 41
Table 17: Chamber comparison of the ….. : mean, SD, and CV% for ControlArt (CA) and Ascott (AS) per material (n = 6, both orientations pooled). * CV% for A2202F03 not interpretable. 43
Table 18: Individual position averages (mm) for all materials in both chambers (D pos 1–3 = diagonal positions; H pos 1–3 = horizontal positions). * marks outlying positions driving chamber differences. 44
Table 19: Final material ranking by overall mean corrosion propagation (n = 12 per material, both chambers and orientations combined). 45

List of acronyms and abbreviations

ACT I / II / III: Accelerated Corrosion Test (versions I, II, III)

ANOVA: Analysis of Variance

ASTM: American Society for Testing and Materials

ATR-FTIR: Attenuated Total Reflectance Fourier Transform Infrared Spectroscopy

CETP: Corporate Engineering Test Procedure (Ford)

CT: Computed Tomography

CV%: Coefficient of Variation (percentage)

DAS: Data Acquisition System (mean offset limit ±0.4°C / ±2%RH)

DFT: Density Functional Theory

ECC1: Renault Corrosion Test Standard

GDP: Gross Domestic Product

GI: Galvanised Iron (GI 50/50)

GMW14872: General Motors Worldwide Engineering Standard

IMPACT: International Measures of Prevention, Application, and Economics of Corrosion Technologies (NACE study)

IQR: Interquartile Range

ISO: International Organization for Standardization

MSPDA: Maximum Short-Period Deviation of the Average (limit ±0.6°C / ±3%RH)

NACE: National Association of Corrosion Engineers

NDE: Non-Destructive Evaluation

NSS: Neutral Salt Spray

OEM: Original Equipment Manufacturer

PID: Proportional-Integral-Derivative (control algorithm)

RH: Relative Humidity

RQ1 / RQ2 / RQ3: Research Question 1 / 2 / 3

RSM: Response Surface Methodology

SAE: Society of Automotive Engineers

SCC: Stress Corrosion Cracking

SECM: Scanning Electrochemical Microscopy

SEM: Scanning Electron Microscopy

TC210 / TC220: ControlArt chamber model numbers

VCS: Volvo Car Standard (VCS 1027,1449)

VDA: Verband der Automobilindustrie (German Automotive Industry Association)

WP1–WP5: Work Package 1 through 5

XCT: X-ray Computed Tomography

1 Introduction and Background

This thesis work addresses the problem regarding variability in accelerated corrosion testing, the industrial context at Volvo Cars, the goals of the project, and the overall research methodology. It introduces the two primary commercial test chambers under evaluation: the **ControlArt TC210** and the **Ascott Atmosfär Premium AT1300ip,** and outlines how environmental laboratory testing is used to replicate component life-cycle degradation within a condensed laboratory timeframe.

In the automotive sector specifically, corrosion directly affects the structural integrity, passive safety, and aesthetic quality of vehicles throughout their service life. For a manufacturer such as **Volvo Cars**, understanding and predicting the long-term degradation behaviour of materials is essential at every stage of the development process.

Because real-world corrosion damage takes years or decades to manifest under typical service conditions, the industry relies on accelerated corrosion test chambers to compress this timescale into weeks. These chambers apply cyclic sequences of salt spray, temperature variation, and controlled humidity that intensify the electrochemical processes governing atmospheric corrosion, enabling the relative performance of materials, coatings, and assembled components to be assessed within a standard laboratory programme. At Volvo Cars, the primary protocol for this purpose is the **ACT II cycle** (Accelerated Corrosion Test, Version II, VCS 1027,1449), which aims to replicate approximately six years of in-service chloride exposure within a six-week test.

A fundamental, but often under-examined, assumption in accelerated corrosion testing is that results obtained in different laboratories or used of different chamber design are comparable. In practice, different chamber designs implement the same programmed test cycle through different engineering solutions: different airflow systems, spray geometries, heating elements, and control algorithms. These hardware differences can produce spatially and quantitatively different corrosive environments even when the nominal cycle parameters are identical. This comparability question has direct industrial consequences: if a coating system is qualified in “Chamber A” at one facility, can a supplier or partner laboratory running “Chamber B” be confident that the same material would achieve the same result? If not, the qualification is hardware-dependent rather than material-dependent, and its value as a predictive tool is fundamentally compromised.

This comparability problem is not limited to comparisons between different laboratories. Even within the same facility, the ageing of chamber hardware, variability in maintenance schedules, and operational malfunctions can cause a chamber to deviate from its specified setpoints. The Ascott Atmosfär Premium AT1300ip used in this project experienced a temperature malfunction during the test campaign, failing to consistently reach the 50 °C drying-phase setpoint of the ACT II protocol, and other issues addressed below. Rather than treating this as a disqualifying experimental failure, this master thesis treats it as a direct and concrete case study in the broader question: how sensitive are ACT II test outcomes to deviations from the protocol's environmental setpoints, and do the standards' tolerance bands provide sufficient protection against such deviations?

The answer to this question has implications not only for the interpretation of results from this specific study, but also for the broader framework of how corrosion test standards are written and how chamber calibration requirements are specified. If the Ascott chamber, operating outside its temperature specification, still generates results comparable to the correctly operating ControlArt chamber, the current standard may permit more hardware variability than necessary for meaningful comparability. If it does not, the standard's requirements are justified, and the importance of tight calibration is confirmed. Either finding advances the field.

There is also a forward-looking dimension to this work. The automotive industry is increasingly moving towards testing of larger and more complex assembled components in “walk-in” corrosion chambers, room-sized enclosures capable of accommodating full vehicle sub-systems, or even complete body-in-white structures. A new test cycle, informally referred to as **ACT III**, is under development for this class of large-

component testing. Before such a cycle can be standardised, the fundamental question of chamber-to-chamber comparability must be resolved for the smaller, better-characterised ACT II format. The methodology developed in this master thesis: a structured sample-based route for comparing chambers and validation of their ability to deliver a given test cycle, is therefore intended to provide a foundation that could eventually be extended to the ACT III context. This extension is explicitly out of scope for the present work.

1.1 Problem

Despite the use of standardised protocols, the outcome of an accelerated corrosion test is shaped not only by the programmed cycle but by the engineering design of the chamber executing it. Differences in temperature uniformity, airflow distribution, spray geometry, and the control accuracy of phase transitions can produce significantly different corrosive conditions across the test volume and between different chamber designs. Currently, there is no established and validated routine method for determining whether a given chamber can deliver ACT II conditions with sufficient fidelity to produce comparable results.

The core research questions addressed in this thesis are:

1. ***Chamber equivalence****: Can a comparative methodology using a standardized set of samples be developed to determine whether a new chamber can follow the Volvo standard corrosion cycles?*
2. ***Standard requirements****: Given the Ascott chamber temperature malfunction, does the result fall within a range where they remain comparable with a functioning ACT II chamber (Control Art)?*
3. ***Scope for standard improvement****: Which aspects of the current standard's calibration and acceptance criteria lack sufficient specificity, and what modifications or stricter requirements could help reduce inter-laboratory variability?*

1.2 Purpose

The purpose of this master thesis was to evaluate how the design and operational state of an accelerated corrosion chamber influences ACT II test outcomes, and to develop a comparative methodology that can determine whether different hardware platforms deliver equivalent test conditions. The primary industrial beneficiary is **Volvo Cars**, specifically the Material Centre, which requires high-confidence and hardware-independent corrosion data to approve material combinations and joining methods for production vehicles. By establishing that results from the ControlArt and Ascott platforms can be compared, the engineering teams would gain the ability to use whichever chamber is available without introducing systematic uncertainty into the approval process.

The Ascott chamber temperature malfunction introduces a second dimension of purpose: to use the deviation from the standard's setpoints as a natural experiment for assessing standard sensitivity. If a chamber that fails to meet the drying-phase temperature setpoint still produces comparable corrosion results for certain material types, this provides direct evidence that the standard's temperature tolerance for those materials may be broader than currently required. If it does not, the malfunction data quantifies the magnitude of the bias introduced by the deviation.

Another purpose of the study was to lay the groundwork for a **chamber qualification workflow**: a structured, sample-based procedure that any chamber supplier could apply to assess whether a chamber model is ready to run ACT II tests. While this workflow is developed and demonstrated in the context of the ControlArt/Ascott chamber comparison, it is designed to be generally applicable to other chamber suppliers. In the longer term, a validated version of this workflow could serve as the basis for qualification procedures for the next generation of large-component test systems, including "walk-in" chambers and the ACT III cycle currently under development.

Sustainability and Ethical Considerations: This work contributes directly to **SDG 9 (Industry, Innovation and Infrastructure)**: a more reproducible qualification methodology reduces redundant test repetitions across laboratories, lowering energy and chemical consumption. It also supports **SDG 12 (Responsible Consumption and Production**) by enabling more precise material dimensioning; components correctly qualified are neither under-specified nor over-engineered, reducing material waste and vehicle weight over the service life.

1.3 Goals

The primary aim of this master thesis project was to investigate how the design and operational behaviour of the ControlArt TC210 and the Ascott Atmosfär Premium AT1300ip influence ACT II test outcomes, to assess whether the Ascott is capable of performing the ACT II cycle reliably, to evaluate the adequacy of the standard's tolerance bands, and to develop a reusable workflow for comparing and qualifying corrosion chambers. This primary aim was structured into the following sub-goals or also called work packages:

Sub-goal 1 Assess Ascott capability for ACT II: Determine whether the Ascott AT1300ip chamber, including under the conditions created by its temperature malfunction, is capable of producing ACT II test results equivalent to those of the correctly operating ControlArt TC210 chamber.

Sub-goal 2 Chamber homogeneity characterisation: Characterise the spatial distribution of corrosion severity across the full test area of both chambers using zinc foils samples, providing a quantitative map of within-chamber uniformity for both total mass loss and localised pitting.

Sub-goal 3 Scribed panel corrosion propagation: Evaluate the lateral corrosion propagation from standardised artificial scribes on painted panels of four material systems across multiple positions in each chamber, assessing both material ranking consistency and position-to-position reproducibility.

Sub-goal 4 Crevice corrosion: Investigate the homogeneity of corrosion within crevice geometries.

Sub-goal 5 Complex component testing: Evaluate the corrosion behaviour of real automotive sub-assemblies incorporating sealing plans, dissimilar material combinations, and different joining methods.

Sub-goal 6 Chamber qualification workflow and standard assessment: Develop a structured, sample-based workflow for qualifying chambers and assessing their capability to perform the ACT II cycle. Identify specific aspects of the current ACT II standard where the calibration requirements are insufficiently detailed and then propose targeted improvements to reduce inter-laboratory variability.

The project delivers three primary outputs:

(1) a **comparative method** based on structured indicator testing to align and assess results between chamber hardware;

(2) a **chamber qualification workflow** providing a reusable procedure for validating chamber readiness before a test campaign; and

(3) a quantitative **assessment of ACT II standard adequacy**, identifying where more firm, or specific calibration requirements would improve cross-laboratory reproducibility.

1.4 Research Methodology

This master thesis adopts an **experimental and comparative approach** as its core research methodology. The central question, whether two different corrosion chamber designs deliver equivalent ACT II test outcomes, is empirical that can only be answered through direct measurement under controlled and reproducible conditions. Neither computational modelling nor literature review alone can provide the concrete evidence required.

The study assumes that a real, objective difference in chamber behaviour either exists or does not, and that this difference can be measured and quantified to a sufficient degree of confidence through systematic experimental observation. The measurements themselves are treated as empirical approximations of that underlying reality, subject to quantified uncertainty but not fundamentally subjective.

The research is further structured as a **multi-method study**: a single primary methodology (experimental comparison) is combined with several measurement techniques such as gravimetric analysis, optical microscopy, CT scanning, and statistical analysis. This multi-method design reduces the risk that any single measurement artefact drives the conclusions, since different techniques must produce a coherent picture for a finding to be considered robust.

The **Ascott temperature malfunction** that occurred during the first exposure campaign is treated within this methodological framework as a *controlled perturbation* rather than a confounding failure. The deviation from the ACT II temperature setpoint is fully documented through the chamber's logged operational data, making it a bounded and characterised experimental variable. By comparing results from the malfunctioning Ascott chamber against the correctly operating ControlArt chamber, the study generates direct empirical evidence about the sensitivity of ACT II test. This reframing is methodologically valid because the malfunction did not invalidate the data. It changed the question the data can answer, from a clean chamber-design comparison to a chamber-design comparison that also yields information about standard robustness.

The project was structured into five Work Packages (WP1–WP5) that progress in complexity from fundamental environmental characterisation of the chambers to the testing of real automotive components.

1.5 Delimitations

The following boundaries apply explicitly to this master thesis project:

- The study evaluates only the **ACT II cyclic protocol** (VCS 1027,1449 / CETP 00.00-L-467). Other protocols that can be run on the same hardware, including ACT I, are not evaluated.
- Electrochemical measurement techniques (electrochemical impedance spectroscopy, potentiodynamic polarisation) are not used. Corrosion behaviour is characterised entirely through post-exposure physical, dimensional, and corrosion product analysis.
- The root cause of the Ascott chamber temperature malfunction is not investigated at the software and hardware levels. The thermal deviation is documented from logged chamber data and treated as a bounded experimental variable.
- CT scanning is applied to selected components only, not to all specimens across all work packages, due to instrument time constraints.
- The extension of the chamber comparison workflow to "*walk-in" chambers* and to the *ACT III cycle* for large automotive components is identified as future work and is not examined. The methodology developed here is designed with this extension in mind, but its applicability to those contexts remains to be validated.

2 Background

Corrosion is fundamentally defined as an electrochemical reaction between a metal and its surrounding environment that leads to the degradation of the material and its properties [5]. It is a spontaneous process driven by thermodynamic instability. Most metals, when exposed to natural environments, tend to transform and stay in low-energy and thermodynamically favourable states in chemical compounds as oxides, hydroxides, and sulphides [1]. This chemical transformation is carried out through chemical phenomena such as corrosion.

Beyond material loss, corrosion poses significant risks to structural integrity, economic stability, and environmental safety across virtually every engineering sector, from the petrochemical and offshore energy industries to civil infrastructure, biomedical devices, and the automotive industry [2].

The economic toll of corrosion is impressive. The NACE International IMPACT (International Measures of Prevention, Application, and Economics of Corrosion Technologies) study of 2016 (the most comprehensive global assessment to date) estimated direct corrosion costs at approximately US$2.5 trillion per annum, equivalent to 3.4% of world GDP [3]. Iannuzzi and Frankel (2022) subsequently updated this picture: when indirect costs are accounted for alongside direct ones, the total cost of corrosion may exceed 6.2% of global GDP, amounting to roughly US$6 trillion based on 2021 World Bank GDP figures [4]. The 2016 IMPACT study also estimated that deploying existing and well-proven corrosion control practices could realise savings of between 14% and 35% of the direct cost, representing hundreds of billions of dollars of preventable expenditure every year [3].

Beyond the financial dimension, corrosion carries an overlooked environmental burden. Iannuzzi and Frankel (2022) provided the first quantitative estimate of the carbon footprint of steel corrosion, determining that the CO_2 emissions associated with producing replacement steel for corroded structures will account for 4.1–9.1% of total EU and US greenhouse gas reduction targets by 2030 [4]. This underscores the urgency of sustainable corrosion management both as an economic- and a climate imperative. This theoretical background will explain the fundamental electrochemical principles governing corrosion, its variables, and the principal typologies encountered in automotive engineering practice.

2.1 Fundamentals of Corrosion

Electrochemical Nature of Corrosion

Corrosion of metals in the presence of moisture (as in water) is inherently electrochemical in character. Four essential components must be present simultaneously for it to occur: an **anode**, a **cathode**, an **electrolyte** connecting the two electrodes, and a **conductive path** (here a metal) permitting the flow of electrons between anode and cathode [5]. The removal of any one of these components is sufficient to arrest the corrosion process, which is why many practical protection strategies are based on removing one or more of these components.

At the anode, an oxidation reaction takes place in which metal atoms give in electrons that pass into the electrolyte as positively charged ions. For steel, the primary anodic reaction is expressed as:

$$Fe \rightarrow Fe^{2+} + 2e^-$$

At the cathode, the electrons coming from the anode are consumed by a reduction reaction. In aerated, near-neutral to alkaline environments, oxygen reduction is the dominant cathodic reaction:

$$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$$

In acidic environments, hydrogen evolution replaces or supplements oxygen reduction as the cathodic reaction:

$$2H^+ + 2e^- \rightarrow H_2$$

These coupled anodic and cathodic half-reactions constitute the corrosion cell. The separation of anodic and cathodic sites may occur at the microscale (e.g., between different microstructural phases within a single alloy) or at the macroscale (e.g., contact pads and brake discs exposed to moisture) [5]. The electrolyte, which may be aqueous solution, moisture, or any other ionically conductive medium, facilitates the migration of ionic species between these sites, closing the electrochemical circuit [6].

Thermodynamics of Corrosion

The thermodynamic feasibility of any corrosion reaction is governed by the Gibbs free energy change (ΔG) of the overall cell reaction. Spontaneous corrosion occurs when ΔG < 0, a condition equivalent to the electrochemical cell having a positive open-circuit potential difference (E > 0). The relationship between cell potential and free energy is described by the following equation:

$$\Delta G = -nFE$$

where n is the number of electrons transferred and F is Faraday's constant (96,485 C mol^{-1}). The equilibrium electrode potential for any half-reaction as a function of ionic activity and temperature is described by the Nernst equation [7], useful for predicting the driving force of corrosion under specific chemical conditions. Another useful tool is the Pourbaix diagram (a potential–pH stability diagram) that describes only thermodynamic equilibrium conditions but cannot predict the actual rate of corrosion (i.e. the kinetics).

Kinetics of Corrosion

Thermodynamics establishes whether corrosion is possible, but kinetics governs the rate at which it actually proceeds. The kinetics of electrode reactions are described by the Butler–Volmer equation $j = j_0 \left[\exp\left(\frac{\alpha_a nF\eta}{RT}\right) - \exp\left(-\frac{\alpha_c nF\eta}{RT}\right)\right]$, which relates the net current density at an electrode to the overpotential [7]. At sufficiently large anodic or cathodic overpotentials, the Butler–Volmer relationship simplifies to the Tafel approximation: $\eta = \pm A \cdot \log_{10}\left(\frac{i}{i_0}\right)$, predicting a linear relationship between the logarithm of current density and overpotential. The intersection of the anodic and cathodic Tafel lines in a mixed-potential diagram defines the corrosion potential and the corresponding corrosion current density, which can be converted to a corrosion rate via Faraday's law [8].

Passivation

A critical kinetic phenomenon encountered with many metals and alloys is passivation: the formation of a thin, adherent, and highly protective oxide or hydroxide film on the metal surface that dramatically reduces the corrosion rate, often by several orders of magnitude [9]. For stainless steels and aluminium alloys, passivity arises from the spontaneous formation of a chromium-enriched (trivalent Cr^{3+}, Cr_2O_3) or aluminium oxide (Al_2O_3) surface layer, respectively. The passive film acts as a barrier, preventing contact between the underlying alloy substrate and the corrosive environment.

2.2 Factors Influencing Corrosion Rates

The rate at which a metal/alloy degrades is determined by a complex interplay of material properties, geometric factors, and environmental variables. Understanding these variables individually and collectively is essential for accurate corrosion prediction and effective control. Some of the most important factors are described below:

Temperature

Temperature has a dominant influence on the corrosion rate by affecting the kinetics of both electrochemical and chemical reactions. According to the Arrhenius relationship, reaction rates generally

increase exponentially with temperature. For many aqueous corrosion systems, it is commonly observed that the corrosion rate approximately doubles for every 10 °C rise in temperature, particularly during the initial electrochemical phase [10]. This relationship is not universally monotonic. In closed aqueous systems, elevated temperatures can simultaneously reduce the solubility of dissolved oxygen. As a result, there may exist an intermediate temperature (often in the range of 60–80 °C) at which the net corrosion rate passes through a maximum before declining [10].

pH

The pH of the aqueous environment is one of the most critical parameters controlling the stability and solubility of corrosion products and the kinetics of cathodic reactions. A lower pH accelerates corrosion in two distinct ways: it promotes the hydrogen evolution cathodic reaction ($2H^+ + 2e^- \rightarrow H_2$), and it destabilises protective oxide or hydroxide layers that would otherwise show slow anodic dissolution [10, 11]. For carbon steel, the metal remains active and corrodes rapidly at pH below approximately 4. Between pH 4 and pH 10, its corrosion rates are less strongly pH-dependent and largely governed by the rate of oxygen diffusion to the iron surface. Above approximately pH 10, elevated hydroxide ion concentrations promote the formation of ferrous hydroxide and magnetite films that offer significant protection [11]. This is important mainly for ACT I as this Volvo standards use an acidic solution (not studied within the framework of this master thesis).

Salinity and Chloride Ion Concentration

High salinity increases the ionic conductivity of the electrolyte, facilitating the flow of corrosion current between anodic and cathodic sites [13, 14]. In marine environments, seawater contains approximately 35 g/L of dissolved salts dominated by sodium chloride, making it among the most corrosive natural aqueous environments for carbon and low-alloy steels. Beyond its conductivity-enhancing role, the chloride ion (Cl^-) exerts a specifically deleterious chemical effect on passivating alloys, competitively adsorbing onto oxide surfaces, and facilitating local passive film breakdown at defect sites, initiating pitting corrosion [14].

Metallurgical Factors

The microstructure, composition, and heat treatment state of a metal or alloy profoundly influence its corrosion behaviour. Galvanic coupling between different microstructural phases can establish local corrosion cells at the microscale [15]. Alloying additions that promote passivation, such as chromium (Cr) in stainless steels (requiring a minimum of approximately 10.5 wt% Cr), molybdenum (Mo) for improved pitting resistance, and nickel (Ni) for enhanced stability in alkaline environments. These are the primary metallurgical tools for producing corrosion-resistant alloys [15]. Recent computational approaches, including density functional theory (DFT) calculations and molecular dynamics simulations, have enabled researchers to predict corrosion behaviour of candidate alloy compositions from first principles [16].

2.3 Principal Types of Corrosion

Corrosion manifests in a variety of morphological forms, each with different implications for detection, remaining life assessment, and mitigation strategy. Relevant corrosion types for this master thesis project are shortly described below.

Uniform (General) Corrosion

Uniform corrosion is characterised by approximately even deterioration of the entire exposed metal surface at a relatively consistent rate. It is the most predictable type of corrosion and is possible to quantify through weight loss measurements and electrochemical monitoring [12, 17]. The surface appearance of uniformly corroded steel is characterised by a continuous rust layer which is a mixture of iron oxides and oxyhydroxides, including

goethite (α-FeOOH), lepidocrocite (γ-FeOOH), magnetite (Fe_3O_4), and haematite (α-Fe_2O_3), whose composition and protective qualities depend on prevailing environmental conditions [17].

Pitting Corrosion

Pitting corrosion is a highly localised form of attack resulting in the formation of small cavities penetrating the metal surface. It is widely regarded as one of the most dangerous and insidious forms of corrosion because it can cause sudden, catastrophic structural failures with minimal overall mass loss, making it extremely difficult to detect by conventional weight loss monitoring [2, 17].

According to Frankel (1998), pitting corrosion develops in three successive stages: (1) initiation by passive film breakdown at localised surface defects; (2) growth of metastable pits, which may repassivate before establishing a stable growth regime; and (3) propagation of stable pits in which the local environment becomes self-sustaining and increasingly aggressive [18].

Galvanic Corrosion

Galvanic corrosion arises when two dissimilar metals or alloys are in electrical contact within a common electrolyte. The less noble (more anodic) metal corrodes preferentially and at an accelerated rate [17, 19]. The severity is governed by the potential difference between materials, the ratio of cathodic to anodic surface areas, electrolyte conductivity, and geometric configuration [19].

Crevice Corrosion

Crevice corrosion occurs within narrow, shielded geometrical features where geometry restricts the free exchange of electrolyte with the environment [2, 17]. As metal ions accumulate and hydrolyse, the interior solution becomes increasingly acidic and enriched in chloride ions, creating a self-sustaining aggressive chemistry closely analogous to that within pitting corrosion [17].

Stress Corrosion Cracking (SCC)

Stress corrosion cracking (SCC) results from the synergistic action of a sustained tensile stress and a specific corrosive environment that would not, on its own, cause rapid material degradation [17, 20]. Failure can occur at stress levels far below the material's nominally measured tensile or yield strength, often without prior warning, making SCC among the most dangerous corrosion-related failure modes of corrosion [20]. SCC is highly specific in its material-environment combinations. For example, high-strength aluminium alloys are susceptible in chloride solutions, austenitic stainless steels are vulnerable to chloride-induced SCC at elevated temperatures, and high-strength pipeline steels are at risk in near-neutral pH carbonate/bicarbonate ground solutions [21].

2.4 Accelerated Corrosion Test Chambers

Principles of Accelerated Testing

To assess the long-term corrosion atmospheric performance of metallic surfaces, direct field exposure tests are impractical for product development and quality control purposes. Accelerated corrosion test chambers are then helping in this sense as they are designed to replicate and intensify the principal corrosive degradation mechanisms. They do that in a controlled, reproducible, and compressed timeframe so that the relative corrosion resistance of materials, surface coatings, and finished components can be assessed within weeks [22]. This acceleration is achieved by simultaneously applying corrosive agents (principally chloride-containing salt

solutions), elevated temperatures, controlled humidity, and cyclic exposure patterns that mimic the conditions encountered in real service conditions [23].

The fundamental principle underlying accelerated corrosion testing is electrochemical: the same four-component corrosion cell: anode, cathode, electrolyte, and electron path that drives natural corrosion is sustained and intensified within the chamber enclosure. By controlling the composition, concentration, and deposition rate of the electrolyte (salt spray or rain), the temperature (which governs reaction kinetics), the relative humidity (which determines the thickness and connectivity of the electrolyte film on the specimen surface), and the availability of airflow, a chamber can be programmed to recreate specific corrosive climates with high fidelity and good repeatability [23].

Fundamental Parameters Governing Corrosion Chamber Performance: Temperature control and humidity

Temperature is the most fundamental parameter in a corrosion test chamber. It directly governs the kinetics of the electrochemical corrosion reactions (through the Arrhenius relationship, as discussed in Section 2.4.2) and simultaneously controls the evaporation rate of the electrolyte film on specimen surfaces, meaning it controls how wet the specimen surface is at any given time [22]. Uniformity of temperature across the entire test volume is critical for reproducibility. A temperature gradient within the chamber will result in differences in corrosion rates across the testing area, undermining the statistical validity of the test results. Modern chambers achieve uniformity through carefully engineered airflow patterns, custom heat exchanger geometries, and the use of independent heating elements in different chamber zones [22, 23].

In cyclic corrosion testing, the chamber must also be capable of controlled and rapid transitions between temperature setpoints, for example moving from a high-temperature drying phase to a lower phase, while maintaining the other parameter constant or even modifying them as well as specified by each standard.

Humidity Control

The relative humidity (RH) within the test chamber governs the formation, thickness, and ionic conductivity of the thin electrolyte film that forms on specimen surfaces. At RH values below the critical relative humidity for a given salt system (approximately 75–80% for sodium chloride), no continuous electrolyte film forms and the corrosion rate falls dramatically. Above this threshold, corrosion rates increase sharply as film thickness and connectivity increase [12]. Chambers that cannot control RH precisely will therefore produce inconsistent and irreproducible corrosion patterns.

The humidity setpoint required varies substantially between the phases of a typical cyclic test programme. Wetting phases (condensation humidity) typically target 95–100% RH at elevated temperature to produce condensation on specimen surfaces, mimicking overnight dew formation. Controlled humidity phases may target a specific intermediate RH at a defined temperature, replicating the average daytime atmospheric conditions for a given geographic region. Drying phases target low RH or uncontrolled ambient conditions to allow specimen surfaces to dry, terminating active electrochemical dissolution until the next wetting or salt spray event [23]. Advanced chambers achieve active humidity control through the combination of a humidification system, a refrigeration-based dehumidification system and proportional-integral-derivative (PID) control algorithms [22].

Salt Spray Generation and Distribution

The generation and distribution of the chloride-containing salt spray or salt rain is the primary mechanism by which the corrosive electrolyte is deposited on specimen surfaces. In continuous salt spray tests, a compressed air stream is passed through a salt solution reservoir and forced through a nozzle (atomiser) to produce a fine mist of droplets, which then settle onto the specimen surfaces [23].

In cyclic corrosion tests, the delivery mechanism is often more sophisticated. Oscillating or swaying spray bars, positioned at a high level within the chamber, they distribute the salt solution more uniformly over large specimen areas. The salt solution for ACT II is typically a 0.5±0.05% sodium chloride (NaCl) solution in deionised water, with a conductivity that falls between 7.5 and 9.1 mS/cm. The fall-out rate of the salt spray is a critical parameter specified by the test standard and it's verified by a procedure called the downfall test that measures the quantity of atomised water on the test area [24]. The concentration of the salt solution and the deposition rate are independently controllable parameters that together determine the chloride loading on the specimen surface and, consequently, the severity of the corrosive attack [23].

Airflow Management

Airflow within a corrosion test chamber serves several distinct functions that profoundly influence the corrosion conditions experienced by exposed specimens. First, in drying phases, a controlled flow of ambient or heated air across the specimen surfaces promotes evaporation of the electrolyte film, transitioning the specimen from an active electrochemical state to a dry, non-corroding state. The velocity, temperature, and humidity of this drying air stream determine the drying rate and the final moisture content of any surface deposits, which in turn affect the nature of the corrosion products formed [23]. Second, airflow assists in distributing temperature and humidity uniformly throughout the test volume, suppressing gradients that would otherwise produce spatially variable corrosion patterns across the specimen array.

The design of the airflow distribution system is an important engineering challenge in corrosion chamber design and includes the number, placement, and flow rates of inlet and outlet air streams, and the aerodynamic geometry of the test volume. Poor airflow uniformity is a primary cause of within-chamber variability [24].

Cyclic Test Programme Control

Modern corrosion testing has largely moved from continuous salt spray exposure toward cyclic test protocols that sequentially apply multiple climatic phases in a programmed sequence. A typical automotive cyclic corrosion test (such as VDA 233-102, ISO 16701, or SAE J2334) comprises a repeating weekly cycle that includes salt spray exposure, controlled humidity exposure, and drying phases, with optional sub-zero freezing phases [23]. The drying-rewetting cycle promotes the formation of corrosion product morphologies (rust, zinc patina, paint delamination) that more closely resemble those produced by years of natural atmospheric exposure, because the same electrochemical processes of electrolyte film formation, anodic dissolution, cathodic reduction, and corrosion product precipitation and transformation occur under comparable thermodynamic driving forces [24].

The programming and control of these multi-phase test cycles require sophisticated, reliable, and user-verifiable software. The controller must accurately track elapsed time in each phase, execute transitions between phases (including actuating valves, switching heating/cooling systems, and adjusting airflow), monitor sensor feedback from temperature and humidity probes, and log all operational parameters at defined intervals to provide a verifiable audit trail of the test conditions [22].

The ACT II (Accelerated Corrosion Test, Version II) Cycle

The Accelerated Corrosion Test version II (ACT II), standardised as Volvo VCS 1027,1449, is a cyclic atmospheric corrosion test method developed by Volvo Car Corporation and jointly adopted by Ford Motor Company as CETP 00.00-L-467, making the two standards functionally equivalent [24].

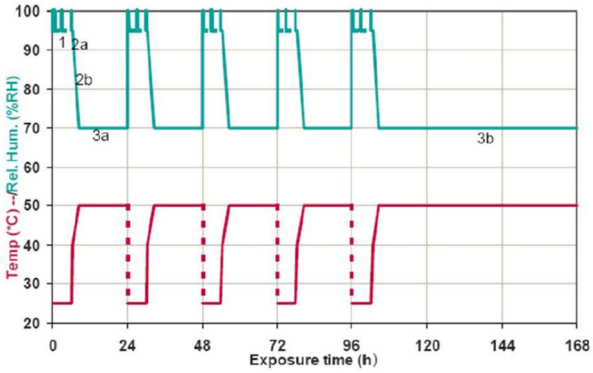


Figure 1: ACT II - Complete weekly cycles in temperature and RH

The complete ACT II test cycle corresponds to one week of calendar time and is composed of two distinct twelve-hour sub-cycles that alternate continuously for the first five days, followed by a 48-hour constant climate phase over the final two days of the week. Sub-cycle 1 begins with six hours of exposure to a 'wet' climate at no less than 95% relative humidity at +25°C, during which the specimens receive intermittent applications of a dilute salt solution (0.5% NaCl) by salt rain. This is followed by a 2.5-hour controlled climate transition, in which the temperature increases from +25 °C to +50 °C while the relative humidity decreases from 95% to 70%. Sub-cycle 2 consists of 15.5 hours of exposure to a constant climate of +50 °C and 70% RH, representing the warm, moderately humid drying phase that allows corrosion products to consolidate and partially dehydrate before the next wetting event. Days six and seven of the weekly cycle comprise a continuous 48-hour exposure at +50 °C and 70% RH, representing an extended drying and ageing period [24].

The 0.5% NaCl concentration of the applied salt rain is lower than the 5% concentration used in continuous neutral salt spray (NSS) tests, reflecting the lower but more realistic chloride loadings encountered in road splash [23, 27]. The test duration is variable depending on the component and acceptance criteria, with typical programmes running from several weeks to several months of cumulative test cycles.

2.5 Comparison of Test Chambers: ControlArt TC210 and Ascott Atmosfär Premium AT1300ip

ControlArt TC210

The ControlArt TC210 chamber is a cyclic corrosion test chamber developed primarily to fulfil the requirements of Scandinavian automotive OEM test programmes, particularly those defined by the Volvo group, and has subsequently been adapted to cover a broader range of international automotive and industrial test standards.

The TC210 employs a flat-bed exposure chamber concept, in which test panels are arranged horizontally across a 2 m² active test area [28].

The 'rain solution' (sodium chloride electrolyte used in salt rain phases) is automatically mixed by the chamber itself, eliminating a potential source of operator error in solution preparation [28].

Ascott Atmosfär Premium AT1300ip

The Ascott Atmosfär Premium AT1300ip is a floor-standing, cyclic corrosion chamber manufactured by Ascott Analytical Equipment Ltd[29]. This brand is rapidly ascending due to the simplicity of their products and relatively lower cost compared to other manufacturers.

Table 1: Parameter comparison: ControlArt TC210 vs Ascott Atmosfär Premium AT1300ip.

Parameter	ControlArt TC210	Ascott Atmosfär Premium AT1300ip
Design concept	Flat-bed cyclic corrosion chamber with horizontal exposure area; modular climate sections.	Floor-standing chamber with vertically oriented test area; oscillating spray bar system.
Dimensions (approx.)	~2 m² exposure footprint; specific external dims not publicly stated	W 2025 × D 1316 × H 1965 mm (1300 L); separate refrigeration unit alongside
Salt spray system	Salt rain and (TC220) salt mist per ISO 9227; automatic rain solution mixing; in-line heaters for mist temperature.	Dual system: oscillating high-level spray bar (5–10 L/m²/h) + conventional atomiser (0.5–5.5 mL/80 cm²/h); peristaltic pumping.
Air management	Scrubber for used air purification; blowing air to remove residual salt mist.	Vertical high-velocity air drying mode.
Test standards (key)	Volvo VCS 1027,149 (ACT I/II); Ford CEPT 00.00-L-467; ISO 16701; Renault ECC1; VDA 233-102	100+ standards: ISO 9227, ASTM B117, VDA 233-102, GMW14872, ISO 16701, SAE J2334, Volvo ACT I/II, Scania, Toyota, Nissan, Hyundai, Ford, Renault ECC-1, etc.
Salt solution system	Automatic mixing of rain solution; internal tank	Separate 110 L reservoir.
Construction	Dedicated flat-bed design; custom heat exchanger parts in exposure chamber	Glass-reinforced plastic, polypropylene & ABS.

Comparative Analysis

From the comparative analysis in Table 1, several key design and capability differences emerge. ControlArt TC210 is fundamentally a capacity-optimised, precision flat-bed cyclic corrosion chamber with a strong heritage in Scandinavian automotive OEM testing. Its principal strengths are the very high specimen throughput, the high uniformity of climatic conditions across the flat 2 m² test area, and the simplicity and low maintenance burden of its design.

Ascott Atmosfär Premium AT1300ip, by contrast, is designed as a universal, multi-standard corrosion testing platform with the widest possible test standard coverage as its primary design objective.

From a fundamental operating principle perspective, both chambers implement the same electrochemical corrosion acceleration strategy: chloride deposition on specimen surfaces to provide the corrosive electrolyte, temperature control to drive reaction kinetics, humidity control to manage electrolyte film formation, and cyclic alternation between aggressive and mild phases to mimic real service exposure cycles. The principal design differences lie in the engineering implementations of each of these parameters.

3 Methodologies and Methods

3.1 Research Process

The study follows a sequential experimental process organised into five Work Packages. Each work package adds a level of geometric complexity to the specimens under evaluation, progressing from simple indicator tiles to real automotive components. The process within each work package is identical in structure: specimen preparation and pre-test documentation, six to twelve weeks of ACT II exposure with weekly photographic inspection, post-test evaluation using the techniques appropriate to the specimen type, and statistical (when possible) analysis of the resulting data.

Both chambers ran simultaneously with nominally identical ACT II programmes, allowing direct side-by-side comparison of results from specimens positioned at equivalent locations within each chamber. Where possible, multiple specimens of the same type were used at each position to provide a measure of within-position variability. The weekly photographs provide the temporal dimension (the rate of corrosion initiation and propagation) while the post-test measurements provide the quantitative outcome dimension.

3.2 Research Paradigm and Philosophical Position

The study operates within a *post-positivist* paradigm. It assumes that the physical quantities of interest (corrosion product mass, pit depth, propagation distance) have objective values that can be measured with quantifiable uncertainty. The role of the researcher is to design experiments that minimise systematic error, apply consistent measurement protocols, and report results with appropriate statistical context. Conclusions are drawn from the convergence of multiple independent measurements rather than from any single observation, reflecting the recognition that individual measurements are fallible approximations of an underlying reality.

The study uses empirical data to characterise chamber performance and to test the hypothesis that two specific chambers produce equivalent ACT II outcomes. This is an *inductive* approach: observations from specific experiments are used to conclude the general question of chamber equivalence and standard adequacy.

3.3 Sample structure

Specimen Types and Work Package Structure

WP1: Calibration: Temperature and humidity logging at multiple positions within each chamber using calibrated independent and internal sensors. This work package does not involve corrosion specimens; it characterises only the physical environment that all subsequent specimens are exposed to.

WP2: Zinc foil chessboard homogeneity test: 4 cm × 4 cm zinc foil tiles arranged in a chessboard pattern across the full floor area of each chamber (13 tiles in Ascott, 18 in ControlArt). These tiles were measured gravimetrically before and after exposure, and selected tiles cross-sectioned for pit depth measurements by means of optical microscopy and CT scanning. ATR-FTIR Spectroscopy and SEM microscope analyses were partly performed but not used to evaluate the tiles.

WP3: Scribed panels: Painted panels of four material systems (aluminium A2202F03, zinc-magnesium ZM2406F08, hot-dip galvanized H2507F08, bare steel S2304F10) with standardised artificial scribes. Three panels per material per chamber were positioned at multiple heights and orientations, and evaluated by lateral blister propagation measurements from the scribe.

WP4: Crevice specimens: Coated fasteners and lap-joint specimens designed to create controlled crevice geometries representative of automotive assembly conditions were evaluated by post-test sectioning and optical microscopy.

WP5: Complex automotive components: Real automotive sub-assemblies including brake hose specimens and, where time permitted, riveted joints and other multi-material assemblies were evaluated by visual inspection, CT scanning, and post-test sectioning.

Specimen Preparation

All specimens were prepared according to the requirements of VCS 1027,1449 and the relevant Volvo material standards before exposure. Scribed panels were scribed according to VCS 1021,29[30] using a standardised scribing tool to a depth penetrating the coating to the bare substrate. Zinc foil tiles were cut to 4 cm × 4 cm, cleaned, and weighed on a precision balance (accuracy ≥ 0.1 mg) before placement. All specimens were photographed in the pre-test condition to document their initial state.

Specimens were positioned within each chamber according to a standardised placement plan developed for this study. All specimens were mounted on non-metallic open racks to allow free air circulation and drainage as specified in VCS 1027,1449 Section 5.1.2.[27]

Exposure and Weekly Inspection

At the end of each weekly cycle, every specimen was removed briefly for photographic documentation, then returned to the same position within the chamber. The photograph records provided a week-by-week visual history of corrosion progression for each specimen.

3.4 Measurement Methods

The following measurement techniques were applied across the work packages. Each technique was selected because it provides information about a specific aspect of corrosion behaviour that the others cannot supply. Combined they provide a complete characterisation of the corrosion outcome.

Gravimetric Analysis

To assess changes in mass, the zinc foil tiles from WP2 were weighed before and after exposure with all corrosion products intact, before any cleaning step. The amounts of corrosion products were calculated as:

Corrosion product (g) = weight after test (uncleaned) − weight before test

This quantity represents the oxide mass gained by the zinc surface and was used as a direct proxy for total zinc dissolution, independent of any cleaning procedure artefacts. A precision balance with 0.1 mg resolution was used for all measurements. Pre- and post-test weighing was performed under the same ambient conditions to minimise systematic error from temperature or humidity effects on the balance.

Cross-Section Optical Microscopy

For pit depth measurements (WP2), blister propagation measurements (WP3), and measurements of other components, selected specimens were cross-sectioned after exposure. For the zinc foil tiles, each tile was cut in half along its length, and the cut face was mounted in cold-setting resin, ground, and polished. All visible pits on the cross-section were measured individually using calibrated optical microscopy at appropriate magnification. For the scribed panels, the maximum lateral blister distance from the scribe centreline was measured at five positions along the scribe length, and the position average was used as the primary metric. For the brake hose, the thickness layer given by the corrosion product was measured.

Because corrosion was visually homogeneous along the length of each zinc foil tile, the half-tile cross-section was taken as representative of the full tile surface. This assumption was validated by comparing CT scan data with optical cross-section measurements on a subset of tiles.

CT Scanning

X-ray computed tomography (XCT), commonly referred to as a CT scan in the laboratory context, is a non-destructive evaluation (NDE) technique that generates high-resolution, three-dimensional reconstructions of the internal and surface structure of a specimen without requiring cutting, sectioning, or any other preparation that would alter or destroy the sample [29]. The technique operates by acquiring a large number of two-dimensional X-ray transmission images at successive angular positions around the specimen, and then applying a mathematical reconstruction algorithm (typically filtered back-projection or iterative reconstruction) to compute a volumetric density map. In corrosion science, XCT has found increasing application over the past two decades because it uniquely enables the three-dimensional characterisation of pitting morphology, including pit depth, volume, shape, and spatial distribution, on specimens that have undergone accelerated or field exposure testing [29].

The non-destructive nature of XCT is its most strategically important attribute in corrosion research. The same specimen can be scanned at multiple time points during an exposure programme, allowing the temporal evolution of individual corrosion features to be tracked with sub-micron spatial resolution without interference from destructive preparation steps [29]. This enables the direct measurement of pit growth kinetics, the initiation and propagation of cracks at pit bases. In the context of this master thesis study, XCT enabled a direct and unambiguous comparison of the attack severity and morphology produced by the two evaluated chamber designs.

ATR-FTIR Spectroscopy

Attenuated Total Reflectance Fourier Transform Infrared spectroscopy (ATR-FTIR) is a vibrational spectroscopic technique in which an infrared beam undergoes total internal reflection at the interface between a high refractive-index crystal (typically diamond, germanium, or zinc selenide) and the sample placed in contact with it. The evanescent wave that penetrates a few micrometres into the sample is selectively attenuated at frequencies corresponding to the fundamental vibrational modes (stretching and bending) of molecular and ionic species present in the sample, producing a characteristic absorption spectrum that serves as a molecular fingerprint [28]. ATR-FTIR requires minimal or no sample preparation and can be performed directly on powdered corrosion products.

In the context of the present study, ATR-FTIR spectroscopy was employed to characterise the chemical composition of the oxide layers formed on the zinc specimens following ACT II cycling in both chamber designs. The identification and semi-quantification of the oxide phases present provides mechanistic insight into the corrosion pathways activated under each set of chamber conditions, and allows comparison of the nature and protective quality of the corrosion product layers produced by the two chamber designs. This information complements the morphological data obtained by SEM and XCT and the quantitative kinetic data from electrochemical measurements.

Scanning Electrochemical Microscopy and Localised Techniques

Scanning electrochemical microscopy (SECM) is part of a family of spatially resolved electrochemical techniques that map the distribution of electrochemical activity across a heterogeneous metal surface at the micrometer scale [15]. Its integration with correlative microscopy has provided mechanistic insight into corrosion initiation at the nanoscale [15]. This technique was used with zinc coupons but then dismissed as very time-consuming and relatively difficult to perceive all the different species.

Statistical Analysis

A consistent statistical framework was applied to all quantitative datasets. The following statistics are reported throughout the thesis, each serving a specific analytical purpose:

Mean and standard deviation (SD): The primary measures of central tendency and absolute spread. All chamber comparisons are based on means; SD quantifies how far a typical individual measurement deviates from the mean.

Coefficient of variation (CV% = SD / mean × 100): The primary homogeneity metric. CV% is scale-free, allowing comparison of consistency between groups with different absolute corrosion levels. A lower CV% indicates more uniform exposure within a chamber or position group. All statements about which chamber is more consistent refer to CV%.

Interquartile range (IQR = 75th − 25th percentile): A robust measure of spread not sensitive to outlier specimens. If CV% is high but IQR is small, one or two extreme values are responsible for the variability, while most specimens behaved consistently. CV% and IQR together give a more complete picture of homogeneity than either metric alone.

95th percentile (p95): The value below which 95% of observations fall. Captures worst-case behaviour of a measurement.

No formal significance tests (t-tests, ANOVA, Mann-Whitney U) are applied in this study. The sample sizes were insufficient for reliable statistical inference. Reporting a p-value with these sample sizes would give a false impression of statistical rigour. Conclusions are instead based on the magnitude and consistency of observed differences across multiple position groups, and on the physical mechanism that explains them.

3.5 Reliability and Validity

Reliability

Reliability is addressed at different levels. At the *measurement level*, calibrated instruments with traceable accuracy were used for all weighing and dimensional measurements. The balance used for gravimetric analysis has a resolution of 0.1 mg; the optical microscope is calibrated with a traceable stage micrometer at each magnification used. At the *test level*, both chambers were running the same automated ACT II programme from the same start time, minimising the influence of cycle timing differences on the comparison.

The Ascott temperature malfunction introduced a known and bounded source of unreliability in the Ascott data relative to the intended ACT II specification. This is explicitly acknowledged and documented rather than treated as a hidden source of error.

Validity

Internal validity (the degree to which observed differences between chambers can be attributed to the chambers rather than to other variables) was supported by the simultaneous exposure of specimens in both chambers under the same programmed cycle, the use of specimens from the same material batches and preparation runs, and the standardised positioning protocol that places equivalent specimen types at equivalent spatial positions within each chamber.

External validity (the degree to which findings generalise beyond this specific study) is intentionally limited. The results apply specifically to the ControlArt TC210 and Ascott AT1300ip chambers under ACT II conditions at the Volvo Cars Material Centre. Generalisation to other chamber models, other protocols, or other laboratories would require additional experimental validation. The methodology and sample-testing route

developed here are designed to be generalisable, but the specific numerical results are not assumed to transfer directly to other hardware or facility combinations.

3.6 Ethical and Societal Considerations

The direct societal relevance of this work lies in automotive safety and sustainability. Corrosion of safety-critical components such as brake hoses, structural joints, and body panels is a genuine risk to vehicle occupants if not adequately characterised and controlled during the development process. By improving the reliability and cross-laboratory comparability of accelerated corrosion test data, this thesis contributes to the development of more durable and accurately specified vehicles, reducing both the risk of premature component failure and the environmental cost of over-engineering components beyond their actual service requirements.

No ethical concerns arise from the experimental work itself: no human subjects are involved, all materials are standard automotive engineering materials, and all chemical handling (salt solutions, resin mounting) follows standard laboratory safety procedures. The reporting of results from a malfunctioning chamber raises a scientific integrity consideration in particular, the obligation to clearly disclose the malfunction rather than suppress or minimise it. This is addressed throughout the thesis by explicit documentation of the Ascott chamber temperature deviation in every context where Ascott results are reported.

3.7 Research Process Scheme

A schematic of the research performed within the framework of this master thesis is given in Figure. 2.

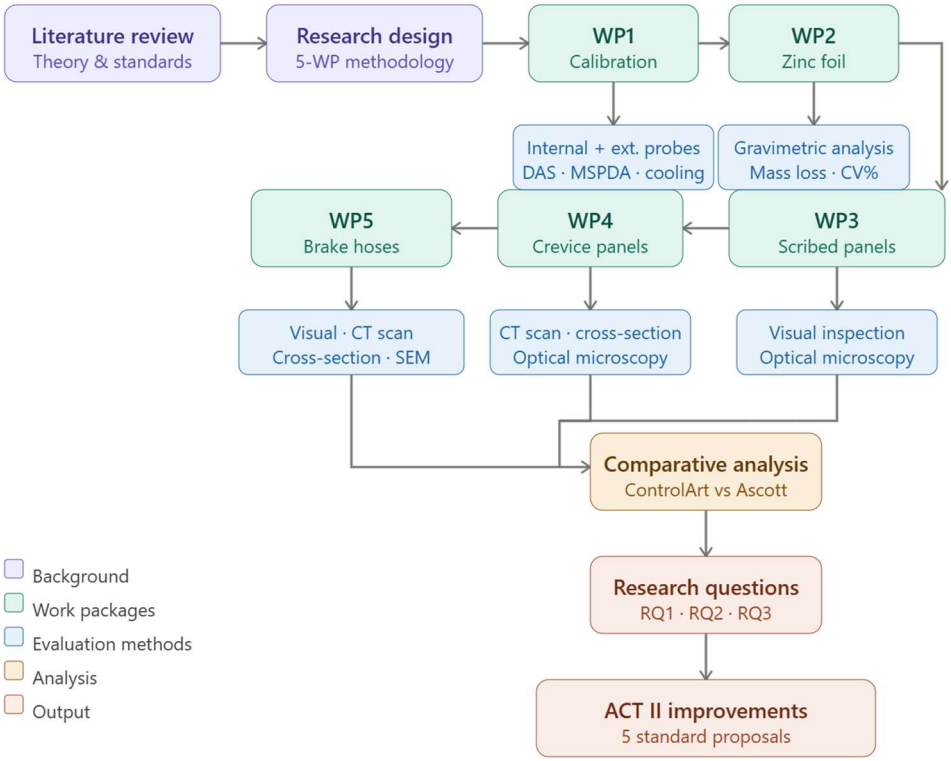


Figure 2: Research Process Scheme

4 Experimental Work: Specimen Description and Preparation

4.1 WP1: Calibration

The environmental characterization was performed using the chambers' internal sensors and two external probes connected to the same independent data logger. One external probe was positioned at the center of the test area, mounted vertically on a panel support to prevent water accumulation on the sensor element. The second probe was placed near the chamber canopy to detect any potential temperature or humidity gradient between the two positions.

The calibration procedure was applied only to the Ascott chamber, which had recently been installed in the laboratory and had not yet been formally validated. The ControlArt chamber was already within its current calibration period and did not require re-calibration at this stage, but it was checked anyways. Both chambers were continuously monitored by their internal sensors throughout the ACT II test campaign to verify that the cycle was running within the specified limits.

The current ACT II standard specifies three climate setpoints that must be verified during calibration: 25 °C/95% RH, 50 °C/70% RH, and 50 °C/95% RH, but does not define a specific test program to do the calibration. The standard requires that the mean value at each setpoint does not deviate from the nominal by more than ±0.4 °C and ±2% RH, and that short-term fluctuations remain within ±0.6 °C and ±3% RH. Since the setpoint combination of 50 °C/95% RH does not occur during the standard ACT II cycle except briefly as a transition, these three conditions cannot be verified by simply monitoring the chamber during a normal test run.

To address this, a dedicated three-step calibration program was designed. Each of the three required setpoints was held for 5 hours, giving a total program duration of 15 hours. The 5-hour duration was chosen to provide a sufficiently long period of stable conditions for reliable evaluation of both mean values and fluctuations.

The first 60 minutes of each step were excluded from the statistical evaluation. This exclusion accounts for the time required by the chamber to ramp between setpoints and reach stable conditions at the new target. During this excluded window, one specific requirement was still verified: the ACT II standard explicitly states that the chamber must cool from 50 °C to 25 °C within one hour.

After the exclusion window, the remaining data from each step were used to calculate the mean temperature and relative humidity and the short-term fluctuations, which were then compared against the acceptance limits of VCS 1027,1449.

4.2 WP2: Zinc tiles

To characterise the spatial uniformity and overall aggressiveness of the two test chambers, a zinc foil chessboard test was conducted in parallel with the main panel exposure campaign. Pure zinc was selected as the indicator material because it corrodes rapidly and predictably under the cyclic conditions of the ACT II protocol, making it an effective integrator of the local corrosive environment over the test duration. Zinc foils were cut to **40 mm × 40 mm** squares. Each tile was cleaned, precision-weighed on a balance before placement, and covered on one face with a resistant adhesive tape, leaving only a **circular area exposed** on the test face. This masking serves two purposes: it produces a clear visual boundary between corroded and virgin material, making weekly progression easier to read, and it ensures that the same exposed geometry is tested at every position across the chamber, eliminating differences in exposed area as a variable.



Figure 3: Zinc tiles samples collection before testing

Tiles were arranged in a chessboard pattern across the full floor area of each chamber. Due to the different sizes of the two instruments, **13 tiles** were placed in a 5×5 grid in the Ascott chamber, and **18 tiles** in a 5×7 grid in the ControlArt chamber. Tile positions were mapped and numbered at the start of the campaign and maintained consistently throughout. Key spatial reference positions were identified as: the four **corners** (TL, TR, BL, BR), the **centre**, and the remaining **inner** tiles. Tiles were exposed to six complete weeks of ACT II cycling. At the end of each weekly cycle, all tiles were removed briefly, individually photographed under standardised lighting conditions, and returned to the same positions within the chamber. This produced a week-by-week photographic record of corrosion initiation and surface progression for each tile and position.

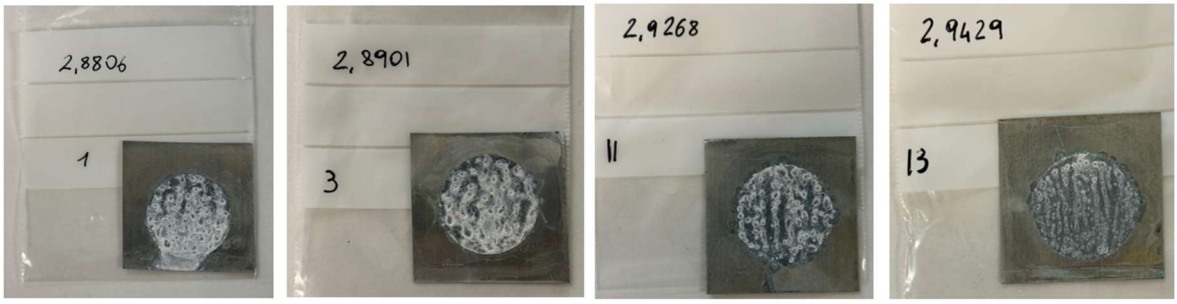


Figure 4: Zinc tiles located at the Ascott's chamber test area corners after a 6-week cycle testing, before deep cleaning.

Each tile was weighed before and after the six-week exposure with all corrosion products intact, i.e. no cleaning was applied before the post-test weighing.

The mass gain represents the oxide mass accumulated on the zinc surface and was used as a proxy for total zinc oxidation per tile position, independent of any cleaning artefacts. After the final weighing, five tiles per chamber were selected for cross-section analysis. Each selected tile was cut in half, the cross-section face mounted in cold-setting resin, ground, and polished. All visible pits on the cross-section were measured individually by optical microscopy to characterise both the depth and distribution of localised attack.

Every two weeks, the same tiles were scanned by X-ray computed tomography (CT). CT scanning was chosen because it is non-destructive. The same tile can be scanned multiple times during the exposure without being opened or altered, allowing the three-dimensional progression of corrosion to be tracked over time. The

machine reconstructs slice-by-slice images and a full 3D model from X-ray measurements taken at many rotation angles. In the resulting images, corrosion products appear as darker, lower-density regions distinguishable from the surrounding metal, and pits appear as dark circular features. To measure pit depth, one face of the tile was used as the origin reference, and the 3D model was navigated inward until no further corrosion was visible, recording the maximum depth reached.

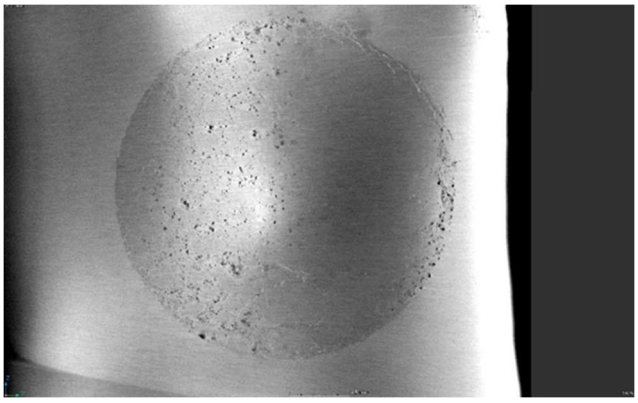


Figure 5: CT scan image of Sample 1 after 2 weeks to the ACT II cycles

Because the tiles are very small and thin and can deform under minimal pressure, flattening and consistent positioning in the scanner were critical to obtaining reliable images.

ASCOTT — 5×5 chessboard (13 tiles)				
N1 Corner TL	N2 Inner			N3 Corner TR
	N4 Inner		N5 Inner	
N6 Inner		N7 CENTRE		N8 Inner
	N9 Inner		N10 Inner	
N11 Corner BL		N12 Inner		N13 Corner BR

CONTROLART — 5×7 chessboard (18 tiles, N14 excluded)						
N14 N14 excl.		N15 Inner		N16 Inner		N17 Corner TR
	N18 Inner		N19 Inner		N20 Inner	
N21 Inner		N22 CENTRE		N23 CENTRE		N24 Inner
	N25 Inner		N26 Inner		N27 Inner	
N28 Corner BL		N29 Inner		N30 Inner		N31 Corner BR

Figure 6: Zinc tiles positioning scheme in both test chambers

4.3 Scribed panels

Four material systems were selected to span the range of corrosion resistance relevant to automotive body and structural components:

- **A2202F03**: aluminium alloy, representative of hood panel material
- **ZM2406F08**: zinc-magnesium alloy-coated steel
- **H2507F08**: hot-dip galvanized steel
- **S2304F10**: uncoated cold-rolled steel, used as the bare substrate reference

Panel preparation and scribing

Panels were prepared and scribed according to VCS 1021,29. Scribing was performed using a hand-held scribing tool equipped with a flat cutting blade of **0.50 ± 0.02 mm edge width** for painted test objects, applied at **90°** to the panel surface. Each scribe was applied in a single continuous pass until the base metal was fully exposed, penetrating through the complete coating stack to the substrate.

Each panel received **two scribed lines**: one **horizontal (H)**, applied at 0 ± 5° from the horizontal, and one **diagonal (D)**, applied at 30 ± 15° from the panel's vertical exposure orientation. The horizontal scribe was positioned below the diagonal, with the lower end of the diagonal in line with or outside the end of the horizontal scribe, as required by the standard. Both scribes were placed at least 10 mm from any panel edge. The recommended scribe length of **70 mm** was used for all panels.

The scribed panels were mounted on non-metallic open racks at **15–20° from the vertical** with the scribed surface facing upward. Three panels per material per chamber were placed at **three different vertical positions** within the chamber, providing spatial replication across the test area height. The panels were exposed for **six complete weeks** of ACT II cycling. Weekly photographic documentation was performed at the end of each cycle.

After six weeks, the panels were cleaned with deionized water before evaluation to improve the visibility of corrosion within and from the scribe without damaging the coating. The corrosion metric recorded was the **one-sided maximum lateral propagation** from the scribe, measured perpendicular to the scribed line. Filiform corrosion was included in the measurement; staining from dissolved corrosion products was excluded. The last 10 mm at each end of the scribe was disregarded in the evaluation. This was done with a graduated lens and then with an optical microscope for better precision.

For each panel, each scribe was assessed at **five equally spaced positions** along its length, recording the maximum lateral blister propagation at each position. The mean of these five readings gives one **position average** per panel per scribe orientation. Using the position average rather than treating each of the five segment readings as an independent observation avoids overstating the effective sample size. This yielded **n = 3 position averages per subgroup** (material × chamber × orientation), giving 48 position averages across the full dataset.

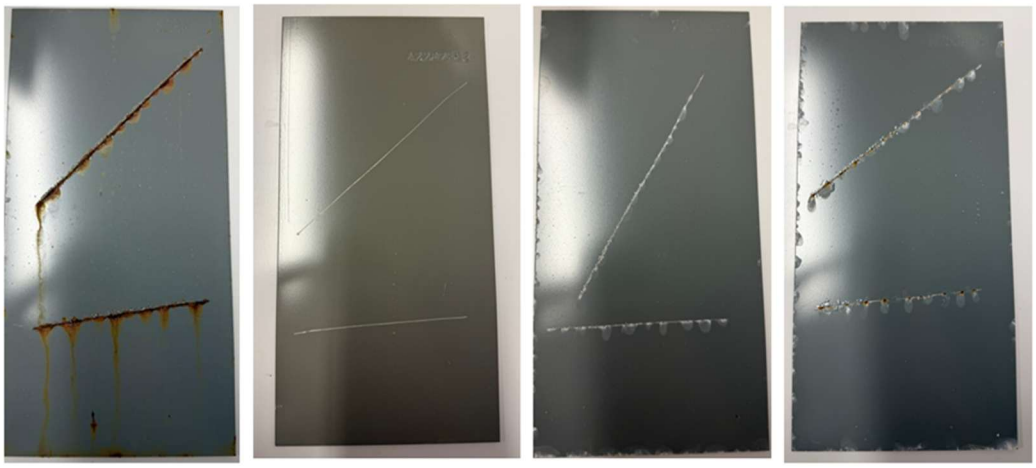


Figure 7: Scribed panels after 4 weeks of ACT II cycle in the ControlArt chamber

4.4 WP4: Crevice Panels

Panels with crevices were assembled by coupling galvanized steel (GI 50/50) with boron steel, separated by a spacer to create a controlled crevice geometry. The two materials were welded together in production to ensure a fixed interface. Then, the samples were split into 2 equal groups: one group underwent the thin film pre-treatment process, and the other group the zinc phosphate pre-treatment process. Following pre-treatment, all panels were painted using a standard automotive paint process. No paint penetration into the crevice was confirmed, meaning the metal-to-metal interface remained unprotected, which reflects real-world joint conditions. A total of 5 replicate panels per treatment were prepared and distributed across the two corrosion chambers in varying positions, for a total exposure duration of 6 weeks under ACT II cycle conditions.

Post-exposure evaluation was carried out in three stages: first, visual inspection of the outer surfaces; second, CT scanning to assess internal corrosion propagation without destructive opening; and third, physical opening of the panels followed by a possible cross-sectioning and optical microscope analysis of the crevice interface. This multi-method approach allowed both macroscopic and microscopic characterization of the corrosion behaviour between the dissimilar metal/alloy substrates.



Figure 8: Example of corrosion of panels with crevices after 6 weeks of ACT2 in the xx chamber

4.5 WP5: Complex components

Brake hose assemblies were selected as representative specimens as a start for more complex geometries. They are responsible for transmitting hydraulic brake fluid from the chassis-mounted hard lines to the brake callipers, enabling the generation of clamping force at the wheel. The metallic elements of the assembly are therefore safety-critical components, subject to both mechanical loading and corrosive attack throughout the vehicle's service life.

The specimens were manufactured from steel then banded and ZiNi coated. Two geometrically distinct variants were included in the test campaign: one specimen with a more gradual curvature along the body of the assembly, and one specimen with a tighter bend radius. The difference in geometry is relevant because changes in curvature affect drainage during the wet phase of the ACT II cycle, local salt accumulation, and the accessibility of internal and recessed surfaces to the spray. ~~To mention also the fact that the more tight the radius is the more the superficial coating might have been ruined during the process.~~ It should also be noted that a smaller bend radius increases the likelihood of damage to the surface coating during the manufacturing process.

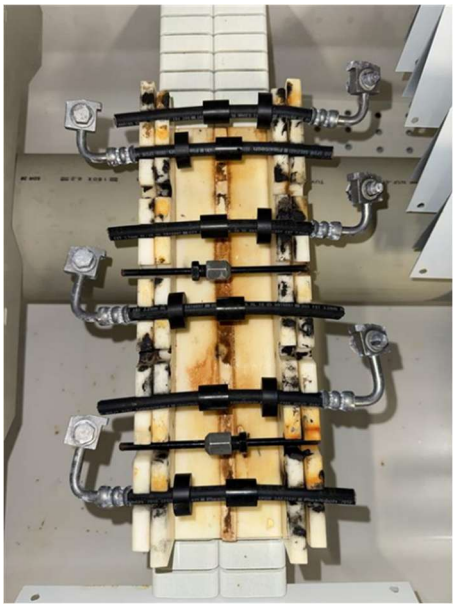


Figure 10 Brake hose sample holder

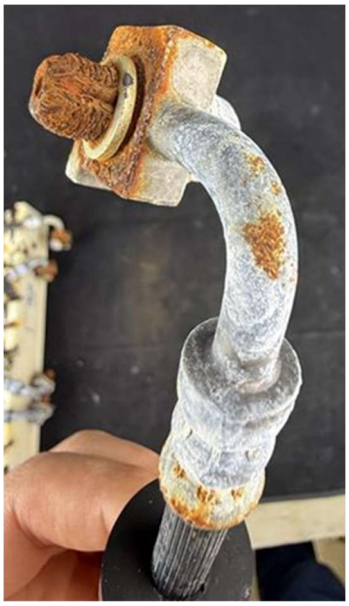


Figure 9 Ascott brake hose after full test

Due to the limited number of available specimens of each variant, no statistically meaningful comparison between the two geometries could be made. The analysis was therefore limited to a qualitative assessment of how corrosion progressed in each chamber over the twelve-week exposure period, rather than a quantitative ranking of the two component types.

The full brake hose assemblies were too long to be placed in the sample holder in their original form. Each assembly was therefore sectioned to retain only the metallic elements of interest: the threaded end fitting connecting to the brake calliper, and the crimped metal ferrule clamping the rubber hose body. The sectioned specimens were mounted in a dedicated sample holder to ensure consistent positioning and orientation throughout the test.

Specimens were exposed for **twelve weeks** of ACT II cycling in both chambers simultaneously. At the end of each weekly cycle, a visual and photographic inspection was performed on each specimen to track the progression of surface corrosion and to identify any anomalies in the corrosion pattern. The inspection protocol was consistent with that applied to all other work packages. In this case there were two different requirements at the second and at the sixth week, no white corrosion and no red corrosion, respectively.

At the end of the six-week campaign, the most corroded sample underwent a **(CT) scanning test** to characterise the three-dimensional distribution of corrosion before any destructive preparation was performed.

Each selected specimen was **cross-sectioned** at the region identified as most affected by corrosion, mounted in cold-setting epoxy resin, and prepared to a metallographic finish by grinding and polishing. The cross-sections were then examined by **optical microscopy** to characterise the depth and morphology of the corrosion layer, the extent of penetration beneath the original surface, and any layered corrosion product structure within the attacked zone. This analysis provides a direct measurement of how deeply corrosion progressed into the material over the test duration, and allows a comparison of the corrosion mode between the two chambers.

5 Results and Analysis

5.1 WP1: Calibration

Prior to the calibration setpoint analysis, the water spray distribution of both chambers was verified by a downfall test. The test was conducted by positioning standardised collection cups across the full test area in a grid layout covering all and running the salt spray system for 30 minutes. The cups were weighed before and after collection, and the measured deposition rate at each position was compared against the acceptance limits of VCS 1027,1449. For the Ascott chamber, all cup positions returned values within the standard limits, and the downfall test was passed on the first attempt. For the ControlArt chamber, two positions initially recorded values outside the acceptable range. The cause was identified as a misalignment of the water atomiser nozzles, which was corrected immediately by tilting the nozzles to redirect the spray distribution. A repeat test confirmed that all positions passed after the adjustment. Both chambers therefore entered the main ACT II exposure campaign with a verified and compliant salt spray distribution.

Ascott AT1300ip The ACT II cycle shown in Figure 10 illustrates the temperature and humidity logs of the Ascott chamber. The temperature data, highlighted in red, demonstrates that the chamber consistently fails to reach the 50 °C setpoint. Consequently, the samples dry later than scheduled, leaving excess moisture and salt on their surfaces. A second temperature-related issue becomes apparent when the cycle is supposed to reach 25 °C. The Ascott chamber exhibits significant inconsistency here.I In most cases, it fails to reach the target temperature, and when it does, it requires more than one hour to stabilize. As a result, the ACT II parameters are not met. Furthermore, because the data fluctuations lack a linear path and exhibit noticeable inhomogeneity, it is evident that the chamber struggles to maintain constant values over time.

This analysis was made before the intervention of the technician and before the problems with the cooling system were fixed.

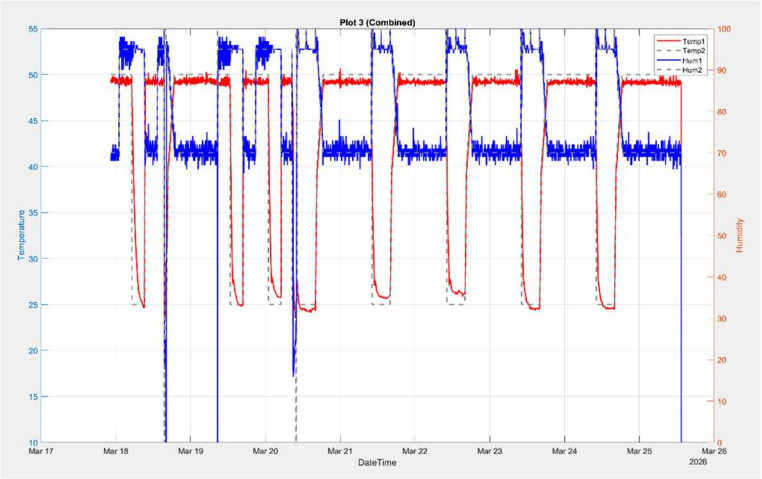


Figure 11: Ascott chamber ACT II cycle log of the temperature and the relative humidity

Figure 12 shows the log of temperature and relative humidity after the chamber was fixed. No issues were registered. I In tables 2 and 3 it is possible to observe thatall mean parameters are inside the standard delimitations.

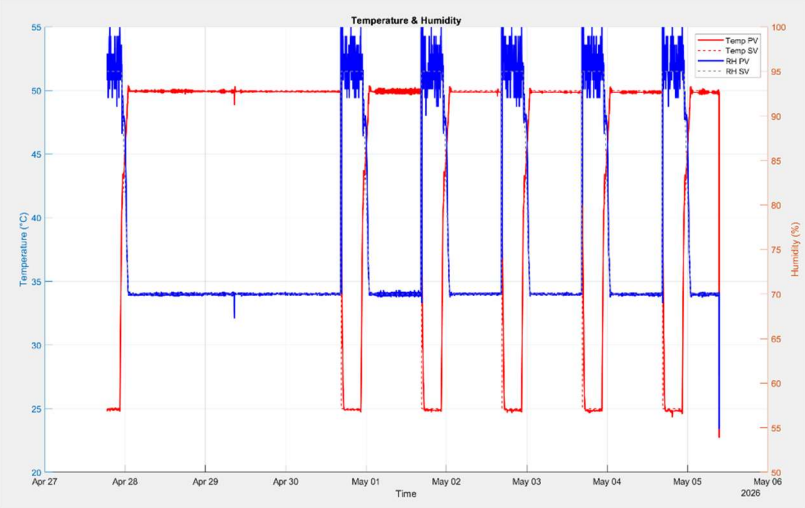


Figure 12 Ascott ACT IIcycle after the first intervention

Table 2: 50 °C and 70% RH setpoints

Start Time	End Time	Duration (h)	Temp Setpoint (°C)	RH Setpoint (%)	Mean Temp (°C)	Mean RH (%)	Temp Deviation	RH Deviation
2026-04-28 00:55:57	2026-04-30 16:25:57	63.5	50	70	49.92	70.00	-0.08	-0.01
2026-05-01 00:55:57	2026-05-01 16:25:57	15.5	50	70	49.93	70.00	-0.07	0.00
2026-05-02 00:55:57	2026-05-02 16:25:57	15.5	50	70	49.88	70.02	-0.12	0.02
2026-05-03 00:55:57	2026-05-03 16:25:57	15.5	50	70	49.86	69.99	-0.14	-0.02
2026-05-04 00:55:57	2026-05-04 16:25:57	15.5	50	70	49.86	69.97	-0.14	-0.03
2026-05-05 00:55:57	2026-05-05 09:29:42	8.5	50	70	49.68	69.89	-0.32	-0.11

Table 3: 25 °C and 95% RH setpoints

Start Time	End Time	Duration (h)	Temp Setpoint (°C)	RH Setpoint (%)	Mean Temp (°C)	Mean RH (%)	Temp Deviation	RH Deviation
2026-04-30 16:37:57	2026-04-30 22:25:57	5.8	25	95	24.91	95.62	-0.09	0.62
2026-05-01 16:37:57	2026-05-01 22:25:57	5.8	25	95	24.88	95.19	-0.12	0.19
2026-05-02 16:37:57	2026-05-02 22:25:57	5.8	25	95	24.87	95.41	-0.13	0.41
2026-05-03 16:37:57	2026-05-03 22:25:57	5.8	25	95	24.84	95.73	-0.16	0.73
2026-05-04 16:37:57	2026-05-04 22:25:57	5.8	25	95	24.82	95.88	-0.18	0.88

Lastly a new attempt of calibration was made, but the chamber could not finish a cycle without giving some critical errors, discovering a new hardware problem with the cooling unit. The chamber in these cases reached temperatures of nearly 80 °C due to some mechanical issues. This event required a new intervention of the technician, and the calibration was repeated (Figure 13).

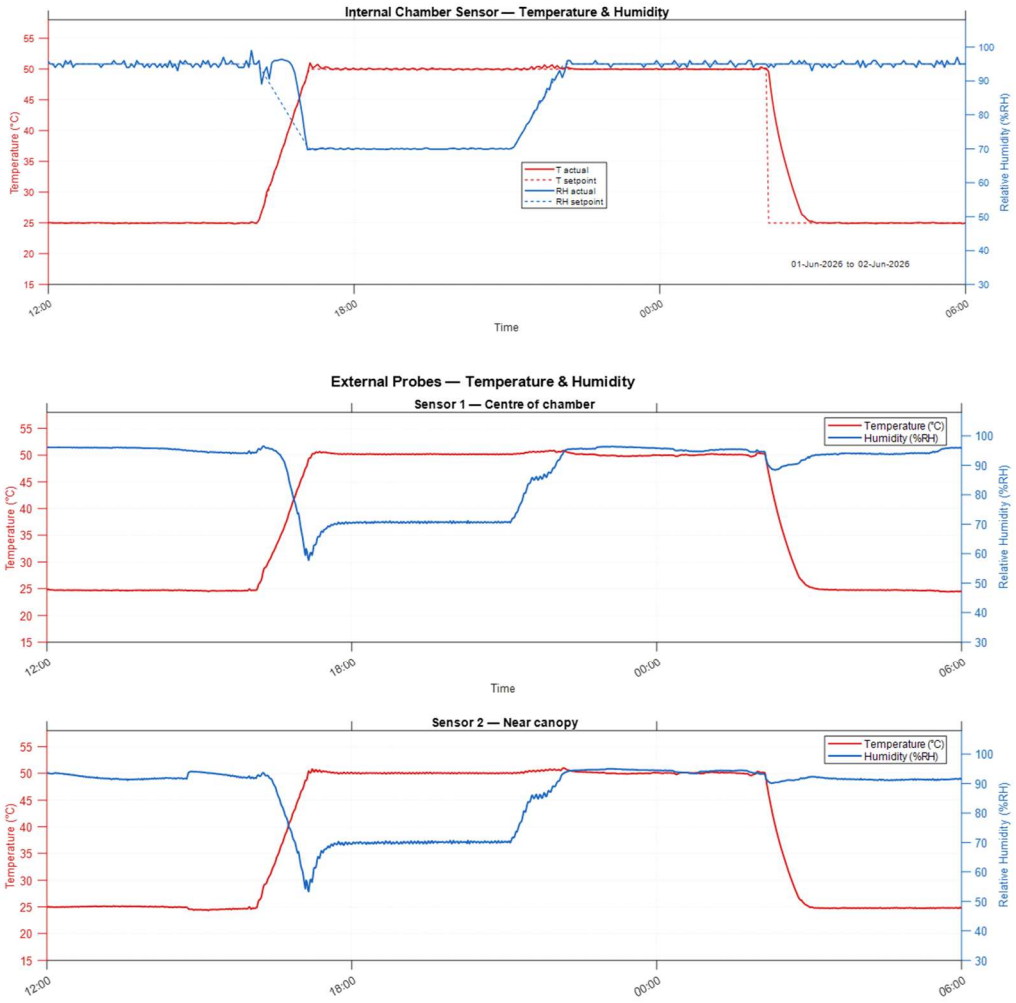


Figure 13: Temperature and relative humidity log from internal and external sensor in the Ascott chamber

Table 2: Ascott AT1300ip — setpoint analysis results for all three sensors and three setpoints. DAS = mean offset from nominal; MSPDA = maximum short-term fluctuation. Limits: DAS ±0.4 °C / ±2% RH, MSPDA ±0.6 °C / ±3% RH.

Setpoint	Sensor		Mean T (°C)	DAS T (°C)	MSPDA T (°C)	Pass T (°C)	Mean RH	DAS RH	MSPDA RH	Pass RH
25°C/95%RH	Internal sensor		24.99	−0.02	0.19	**PASS**	94.98%	−0.02%	4.00%	**FAIL**
	Ext. (centre)	S1	24.67	−0.33	0.59	**PASS**	95.01%	+0.01%	1.50%	**PASS**
	Ext. (canopy)	S2	24.83	−0.18	0.67	**FAIL**	91.92%	−3.08%	4.00%	**FAIL**
50°C/70%RH	Internal sensor		50.05	+0.05	1.01	**FAIL**	69.99%	−0.01%	0.34%	**PASS**
	Ext. (centre)	S1	50.19	+0.19	1.27	**FAIL**	70.00%	+0.00%	12.20%	**FAIL**
	Ext. (canopy)	S2	50.08	+0.08	0.77	**FAIL**	69.23%	−0.77%	16.70%	**FAIL**

50°C/95%RH	Internal sensor		50.00	−0.00	0.35	**PASS**	95.02%	+0.02%	1.00%	**PASS**
	Ext. (centre)	S1	50.00	+0.00	0.43	**PASS**	95.45%	+0.45%	1.40%	**PASS**
	Ext. (canopy)	S2	50.04	+0.04	0.39	**PASS**	94.26%	−0.74%	2.10%	**PASS**

The results of Table 4 is examined and analysed below:

Temperature

All three sensors recorded mean temperatures within the DAS limit of ±0.4°C at all three setpoints, confirming that the chamber was not systematically offset from the nominal temperature (Table 4). The largest mean offset was −0.334°C recorded by Ext. S1 (centre) at the 25 °C/95% RH step, which nevertheless passed the DAS limit easily.

The MSPDA temperature results reveal a more differentiated picture. The internal sensor passed the MSPDA limit at 25 °C/95% RH (0.188 °C) and 50 °C/95% RH (0.354 °C), but failed at 50 °C/70% RH with a value of 1.014 °C, more than 1.7 times the allowed limit. Worth mentioning is that this was the first data available after the one-hour ramping. This is a notable result since the 50 °C/70% RH step is the most demanding step in terms of simultaneous high-temperature and active dehumidification control, and the elevated fluctuation suggests that the PID regulation of the chamber struggles to maintain strict temperature stability while simultaneously driving humidity downward. Ext. S1 (centre) also failed the MSPDA at this step (1.270 °C), while the canopy probe (Ext. S2) failed at both 25 °C/95%RH (0.670 °C) and 50 °C/70% RH (0.770 °C). Conversely, all three sensors passed both DAS and MSPDA temperature limits at the 50 °C/95% RH step, suggesting that simultaneous high-temperature and high-humidity control is less challenging for this chamber than the dehumidification phase.

Relative Humidity

The DAS humidity results are mostly consistent (Table 4). The internal sensor and Ext. S1 (centre) passed the ±2% RH DAS limit at all three setpoints. The most significant finding is the failure of Ext. S2 (canopy) at 25 °C/95% RH, where a DAS of −3.08% RH was recorded. This canopy sensor consistently measured lower humidity than the bulk chamber value at the cold-wet condition, most likely due to its proximity to the chamber walls and the partially insulated plexiglass window, where locally drier and cooler air depresses the measured RH below the nominal setpoint. This spatial non-uniformity was less pronounced at 50 °C, where the chamber's humidification system operates more effectively under higher absolute humidity conditions.

The MSPDA humidity results present a more complex picture across the three setpoints. At 25 °C/95% RH, the internal sensor failed with a value of 4.00% RH, while Ext. S2 also failed (4.00% RH); Ext. S1 passed at 1.50% RH. At 50 °C/70% RH, the MSPDA failures are severe: Ext. S1 recorded 12.20% RH and Ext. S2 recorded 16.70%RH. At 50 °C/95% RH, the performance improved markedly. All three sensors passed both DAS and MSPDA, indicating that the near-saturation condition is the most stable condition maintained of the three setpoints in this chamber.

Cooling Time

Table 3: Ascott AT1300ip, time to cool from 50 °C to 25 °C. VCS 1027,1449 limit: 60 minutes.

Sensor	Time to 25 °C (min)	Result
Internal sensor	57	**PASS**
Ext. S1 (centre)	61	**FAIL**
Ext. S2 (canopy)	52	**PASS**

The internal sensor and Ext. S2 (canopy) both satisfied the 60-minute cooling requirement, recording 57 and 52 minutes, respectively. Ext. S1 (centre), positioned in the middle of the test area furthest from the cooling unit, recorded 61 minutes. Given the 3-minute logging interval of the external probes, the true temperature crossing could have occurred within the preceding minute, placing the actual cooling time between 58 and 61 minutes. The internal sensor confirmed a clear pass at 57 minutes. The Ascott chamber is therefore considered to be at the boundary of the cooling specification, with performance that is conditionally acceptable depending on probe position.

Probe Spatial Gradient

Table 4: Ascott AT1300ip, absolute spatial difference |S2 − S1| over the full calibration run.

Quantity	Mean |Δ|	Max |Δ|
Temperature |T_S2 − T_S1|	0.24 °C	1.62 °C
Humidity |RH_S2 − RH_S1|	2.27% RH	11.10% RH

The mean temperature difference between the two probe positions was 0.2435 °C, indicating acceptable spatial temperature uniformity in steady state. The maximum of 1.620 °C occurred during setpoint transitions, when the heating and cooling system was actively changing conditions and is not representative of steady-state performance. The mean humidity difference of 2.274% RH is more substantial and reflects the canopy-position effect identified in the DAS analysis, where Ext. S2 consistently measured lower RH than Ext. S1. The maximum of 11.1% RH was recorded during the 25 °C/95% RH to 50 °C/70% RH transition, when the chamber was actively dehumidifying and the two probe positions responded at different rates.

Overall Assessment: Ascott AT1300ip

The Ascott AT1300ip partially passed the calibration procedure.

Temperature DAS was adequate for all three sensors at all setpoints, but the MSPDA criterion failed at 50 °C/70% RH for all three sensors, with particularly large exceedances at the external probe positions.

Humidity control was consistent at 50 °C/95% RH across all metrics, but the MSPDA failed at 25 °C/95% RH for the internal sensor and Ext. S2 and failed severely at 50 °C/70% RH for both external probes. The canopy probe failed the humidity DAS at 25 °C/95% RH, revealing a spatial non-uniformity in the cold-wet condition.

Cooling time passed for the internal sensor and the canopy probe, with the centre probe marginally failing.

The chamber, therefore, qualifies conditionally. Its performance at 50 °C/95% RH was fully compliant, but the 50 °C/70% RH dehumidification step remains a point of concern, particularly in the spatial humidity distribution.

Control Art TC210

The ControlArt TC210 calibration data was collected across two separate recording sessions: the 50 °C/70% RH step from a dedicated run exported as VMS_CA_50_70, and the 25 °C/95% RH and 50 °C/95% RH steps from a multi-setpoint run exported as VMS_CA_multi. The Almemo external probe data follows the same two-session structure.

Table 5: ControlArt TC210 — setpoint analysis results for all three sensors and three setpoints. Same limits as Table 5.1.

Setpoint	Sensor	Mean T (°C)	DAS T (°C)	MSPDA T (°C)	Pass T (°C)	Mean RH	DAS RH	MSPDA RH	Pass RH
25°C/95%RH	Internal sensor	25.00	−0.00°C	0.12°C	**PASS**	95.01%	+0.01%	0.55%	**PASS**
	Ext. S1 (M00)	25.09	+0.09°C	0.52°C	**PASS**	94.71%	−0.29%	4.50%	**FAIL**
	Ext. S2 (M01)	25.04	+0.04°C	0.49°C	**PASS**	92.74%	−2.26%	6.10%	**FAIL**
50°C/70%RH	Internal sensor	50.00	+0.00°C	0.11°C	**PASS**	70.00%	+0.00%	1.12%	**PASS**
	Ext. S1 (M00)	49.83	−0.17°C	0.33°C	**PASS**	71.59%	+1.59%	2.90%	**PASS**
	Ext. S2 (M01)	49.75	−0.25°C	0.38°C	**PASS**	70.84%	+0.84%	2.00%	**PASS**
50°C/95%RH	Internal sensor	50.00	−0.00°C	0.15°C	**PASS**	95.00%	+0.00%	0.43%	**PASS**
	Ext. S1 (M00)	49.86	−0.14°C	0.34°C	**PASS**	97.02%	+2.02%	2.10%	**FAIL**
	Ext. S2 (M01)	49.83	−0.17°C	0.47°C	**PASS**	95.41%	+0.41%	0.60%	**PASS**

Temperature

Temperature control in the ControlArt chamber was markedly stricter than in the Ascott chamber across all three setpoints. The internal sensor passed both DAS and MSPDA at all three steps. The most impressive result was observed at 50 °C/70% RH, where the mean temperature was recorded as exactly 50.000 °C (DAS = 0.000 °C) and the MSPDA was only 0.113 °C. At 25 °C/95% RH, the internal sensor achieved a DAS of −0.002 °C and an MSPDA of 0.124 °C, and at 50 °C/95% RH, DAS was −0.002 °C and MSPDA 0.148 °C.

Both external probes passed DAS at all three setpoints. The MSPDA temperature limit was passed by all sensors at all setpoints, with the sole exceptions being Ext. S1 (M00) at 25 °C/95% RH where it was borderline at 0.520 °C, just below the 0.6 °C limit, and no outright failure at any step for temperature MSPDA. This is in marked contrast to the Ascott chamber, where the 50 °C/70% RH step resulted in MSPDA failures across all sensors. The external probes showed consistently negative DAS values at all setpoints (−0.137 to −0.247 °C), suggesting a slight systematic under-reading relative to the internal sensor, likely due to their free-air positioning in the chamber volume rather than wall-mounting (see figures 15 and 16 for the temperature and relative humidity log).

Relative Humidity

The ControlArt internal sensor passed all DAS and MSPDA humidity limits at all three setpoints, a result not achieved by the Ascott. The MSPDA values for the internal sensor were 0.55% RH at 25 °C/95% RH, 1.12% RH at 50 °C/70% RH, and 0.43% RH at 50 °C/95%. This demonstrates fundamentally superior humidity regulation compared to the Ascott, where the internal sensor failed at 25 °C/95% RH (4.00% RH) and at 50 °C/70% RH (3.15% RH, marginally).

The external probes showed notable humidity failures at 25 °C/95% RH. Ext. S2 (M01) failed the DAS limit with a value of −2.26% RH, and both Ext. S1 and Ext. S2 failed the MSPDA limit at this step (4.50% and 6.10% RH respectively). The M01 probe DAS failure at 25 °C/95% RH mirrors the canopy-probe behaviour seen in the Ascott chamber and suggests that positional effects at low temperature and high humidity are a feature of free-air probe positioning in general, rather than a chamber-specific deficiency. At 50 °C/95% RH, Ext. S1 (M00) recorded a DAS of +2.02% RH, marginally exceeding the ±2% RH limit. All other humidity DAS and MSPDA results at 50 °C/70% RH and 50 °C/95% RH passed for both external probes.

Probe Spatial Gradient

Table 6: ControlArt TC210: absolute spatial difference |M01 − M00| over all sessions combined.

Quantity	Mean |Δ|	Max |Δ|
Temperature |T_M01 − T_M00|	0.051 °C	0.14 °C
Humidity |RH_M01 − RH_M00|	1.51% RH	2.20% RH

The spatial uniformity of the ControlArt was consistent. The mean temperature difference between the two probe positions was only 0.0512 °C, and the maximum was 0.140 °C. Unlike the Ascott, where the maximum occurred during transient step changes, the ControlArt's maximum spatial temperature difference remained below 0.15 °C throughout the entire run, including transitions. The mean humidity difference was 1.505% RH, again better than the Ascott's 2.274% RH, with a maximum of only 2.200% RH compared to 11.1% RH in the Ascott, see figure 14. This result is particularly significant given that the ControlArt chamber has a substantially larger internal volume than the Ascott chamber, making spatial uniformity inherently more challenging to achieve.

Control Art TC210 was considered to have passed the calibration.

Comparative Analysis: Ascott vs ControlArt

The environmental characterisation work package produced a clear and differentiated picture of the two chambers' compliance with the VCS 1027,1449 calibration requirements.

The ControlArt TC210 demonstrated strong overall compliance. Its internal sensor passed all DAS and MSPDA criteria at all three setpoints without exception, and its spatial temperature uniformity was particularly impressive. The failures observed were confined to the external probes at 25 °C/95% RH and a marginal DAS exceedance at 50 °C/95% RH, both attributable to the inherent positional sensitivity of free-air probe placement at cold-wet conditions rather than to fundamental chamber deficiencies. The ControlArt TC210 was considered to have passed the calibration and to be fit for purpose as a reference chamber for the ACT II protocol.

The Ascott AT1300ip presented a more nuanced outcome. Following the hardware repairs, the chamber achieved satisfactory temperature and humidity control at 50 °C/95% RH, where all metrics passed for all sensors. However, significant concerns remain at the 50 °C/70% RH dehumidification step, where the internal sensor failed the temperature MSPDA (1.014 °C) and both external probes produced large humidity MSPDA exceedances (12.2% and 16.7% RH). The cold-wet 25 °C/95% RH condition showed humidity MSPDA failures for the internal sensor and the canopy probe, as well as a DAS failure for the canopy probe (−3.08% RH). The cooling time was borderline, with the centre probe recording 61 minutes against the 60-minute specification.

The Ascott AT1300ip is considered to have conditionally passed the calibration. Its performance during the 50 °C/95% RH step was fully compliant, but the 50 °C/70% RH step and the cold-wet condition introduced non-trivial deviations that should be considered when interpreting the corrosion results from subsequent work packages.

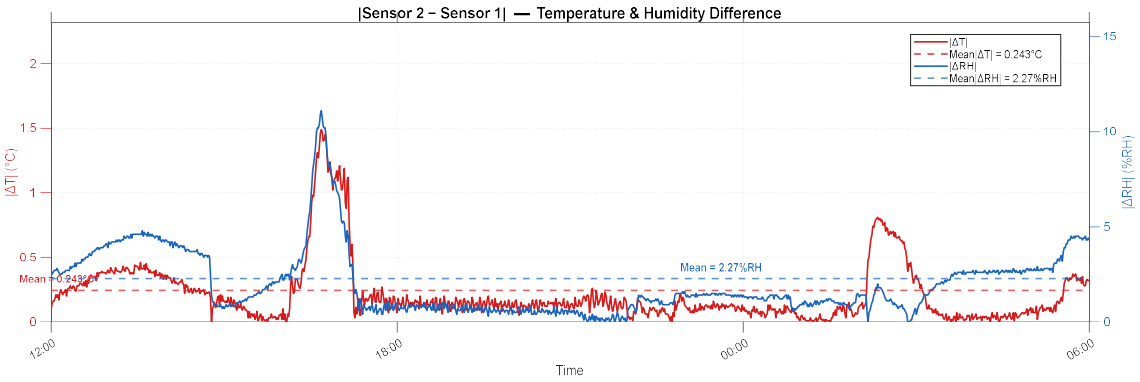


Figure 14: Differences between the two external probes. With Sensor 1 being the central one.

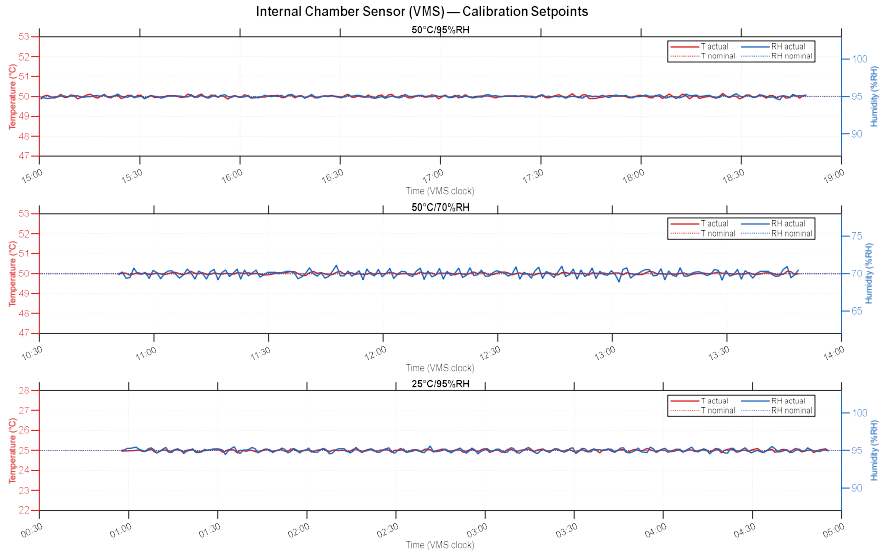


Figure 16: ControlArt internal sensor logs of temperature and humidity. 50°C/95%RH, 50°C/70%RH, 25°C/95%RH

5.2 WP2: Zinc Foil

Mass gain data for the zinc foils is presented in Tables 9 and 10 representative of corrosion product formation for every tile and a first corrosion product analysis sorted by its corresponding position in each chamber.

Table 7: Corrosion product (g) per tile, sorted by position group.

Tile	Position	Corr. product (g)	Tile	Position	Corr. product (g)
N1	**Corner TL**	**0.02920**	**N14**	**Corner TL**	**0.04030**
N3	**Corner TR**	**0.04080**	**N17**	**Corner TR**	**0.00720**
N11	**Corner BL**	**0.02310**	**N28**	**Corner BL**	**0.04150**
N13	**Corner BR**	**0.01150**	**N31**	**Corner BR**	**0.01150**
N7	**Centre**	**0.01550**	**N22**	**Centre**	**0.00790**
N2	Inner	0.06500	**N23**	**Centre**	**0.01970**
N4	Inner	0.05870	**N15**	Inner	0.02430
N5	Inner	0.02240	**N16**	Inner	0.01370
N6	Inner	0.02340	**N18**	Inner	0.02360
N8	Inner	0.02140	**N19**	Inner	0.01470
N9	Inner	0.02530	**N20**	Inner	0.01420
N10	Inner	0.01400	**N21**	Inner	0.01530
N12	Inner	0.03350	**N24**	Inner	0.02460
			N25	Inner	0.01660
			N26	Inner	0.02930
			N27	Inner	0.02260
			N29	Inner	0.01500
			N30	Inner	0.01960

Table 8: Statistical summary of the measured mass of corrosion products listed by the position group and chamber for the zinc tiles. IQR = interquartile range (robust spread measure not affected by outlier tiles).

Group	n	Mean (g)	SD (g)	CV%	IQR (g)	Max (g)	p95 (g)
All tiles: Ascott	13	**0.02952**	0.01640	**55.6%**	0.0121	0.0650	**0.0612**
All tiles: ControlArt	18	**0.02009**	0.00959	**47.7%**	0.0098	0.0415	**0.0405**
Corners: Ascott	4	0.02615	0.01222	46.7%	0.0119	0.0408	0.0391
Corners: ControlArt	4	0.02513	0.01831	72.9%	0.0302	0.0415	0.0413
Centre: Ascott	1	0.01550	—	—	—	0.0155	0.0155
Centre: ControlArt	2	0.01380	0.00834	60.5%	0.0059	0.0197	0.0191

Inner tiles: Ascott	8	0.03296	0.01869	56.7%	0.0177	0.0650	0.0628
Inner tiles: ControlArt	12	0.01946	0.00524	26.9%	0.0089	0.0293	0.0267

Statistical Summary

Table 9: Chamber comparison of the measured mass of corrosion product listed by the position group of exposed zinc tiles. ΔCV% = AS CV% − CA CV%; negative values (pp = percentage points) indicate ControlArt resulted in more variable findings for that group.

Group	AS mean (g)	CA mean (g)	AS−CA (g)	Ratio	AS CV%	CA CV%	ΔCV% (AS−CA)	More consistent
All tiles	0.02952	0.02009	+0.00943	×1.47	55.6%	47.7%	−7.9 pp	**ControlArt**
Corners	0.02615	0.02513	+0.00103	×1.04	46.7%	72.9%	+26.2 pp	**Ascott**
Centre	0.01550	0.01380	+0.00170	×1.12	—	60.5%	—	**Ascott**
Inner tiles	0.03296	0.01946	+0.01350	×1.69	56.7%	26.9%	−29.8 pp	**ControlArt**

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The most significant result in Table 11 is the inner tile comparison. The inner tiles exposed in the ControlArt chamber show a CV% of **26.9%** against 56.7% in the Ascott chamber. The IQR confirms this: CA (ControlArt from now on) inner IQR = 0.0089 g versus AS (Ascott from now on) inner IQR = 0.0177 g. The CA chamber resulted in a substantially more uniform exposure across its 12 inner tile positions than the A chamber across its 8 inner tile positions.

For the corner tiles the picture reverses. The A corner resulted in a CV% (46.7%) well below the C corner, CV% (72.9%). The CA corner scatter is driven primarily by the large gap between N28 (BL corner, 0.0415 g) and N17 (TR corner, 0.0072 g), a factor of nearly six between two tiles in the same chamber at symmetric positions. The A corners ranged from 0.0115 to 0.0408 g, a factor of 3.5. Neither chamber achieves uniform corner exposure, but the CA corner non-uniformity is substantially larger.

Chamber Comparison

The Ascott chamber was more aggressive than the ControlArt chamber in every position group. The overall mean was 0.02952 g versus 0.02009 g, a difference of **+47%** as shown in table 10. This systematic excess is the direct consequence of the temperature malfunction. By failing to reach the 50 °C drying setpoint, the Ascott chamber left the zinc surfaces wet for longer between successive spray cycles. The zinc surfaces remained in continuous electrochemical contact with the chloride electrolyte rather than being periodically dried out. Longer cumulative time of wetness drives more total zinc oxidation.

The consistency comparison confirms that neither chamber is uniformly better across all zones. The **ControlArt chamber showed to be more consistent overall** (ΔCV% = −7.9 percentage points) and dramatically more consistent for the inner tiles (ΔCV% = −29.8 percentage points). The Ascott chamber was more consistent at the corners (ΔCV% = +26.2 percentage points). The inner tile result is the most practically important. The 12 CA inner tiles, which make up the majority of the tile array, were exposed far more uniformly than the equivalent Ascott tiles. The corner result is driven by ControlArt's single outlier pair (N17 vs N28), which substantially inflates its corner CV%.

Corner Homogeneity

Table 10: Comparison of corner-by-corner mass of corrosion product formed after xx days exposure to the ACT II cycle and within-chamber corner homogeneity statistics.

Corner	AS tile	AS corr. product (g)	CA tile	CA corr. product (g)	Difference AS−CA (g)	Higher in
TL corner	**N1**	0.02920	**N14**	0.04030	**−0.01110**	**ControlArt**
TR corner	**N3**	0.04080	**N17**	0.00720	**+0.03360**	**Ascott**
BL corner	**N11**	0.02310	**N28**	0.04150	**−0.01840**	**ControlArt**
BR corner	**N13**	0.01150	**N31**	0.01150	**≈0**	**Equal**

In the Ascott chamber, the highest corner was TR (Top Rigth, N3, 0.0408 g), and in the ControlArt chamber, the highest corner was BL(Bottom Left, N28, 0.0415 g). The outlier is in a different position in each chamber, which argues against a shared geometric cause such as proximity to the spray nozzle inlet. Instead, corner exposure appears to be governed by chamber-specific airflow patterns and drainage geometries that differ between the two instruments. The practical consequence is that *none of the chambers delivers uniform corner exposure*, and that testing at a single corner position would give an unrepresentative result in either chamber.

Pit Depth Analysis

Cross-sections were prepared from five tiles per chamber. In ControlArt: N17 (TR corner), N22 and N23 (two centre tiles), N28 (BL corner), N31 (BR corner). In Ascott: N1 (TL corner), N3 (TR corner), N7 (centre), N11 (BL corner), N13 (BR corner). Tile N14 (ControlArt TL corner) was not examined as the zinc tiles got deformed during hot moulding with the high pressure and temperature. That is also the reason why cold moulding was used for the rest of the samples. Each tile was cut in half, the cross-section face embedded in resin, and all visible pits measured. Because corrosion was visually homogeneous along the tile length, the half-tile section was decided to be representative of the full surface. A total of 368 individual pit measurements were measured for the tiles exposed in the ControlArt chamber. Corresponding number in the Ascott chamber was 391.



Figure 15: Example of cross cut view of a corroded zinc tile exposed for 6 weeks to ACTII at the optical microscope.

Per-Panel Statistics

Table 11 Pit depths and number on zinc tiles exposed. Per-panel pit depth statistics. n = individual pits measured.

Panel	Chamber	Position	n	Mean (µm)	SD (µm)	SE (µm)	CV%	Min (µm)	Max (µm)	p95 (µm)
N17	ControlArt	TR corner	111	**26.07**	13.68	1.30	52.5%	8.36	82.35	**52.02**
N22	ControlArt	Centre	20	**36.23**	19.21	4.30	53.0%	17.14	84.96	**71.50**
N23	ControlArt	Centre	49	**42.58**	15.87	2.27	37.3%	12.08	93.75	**71.41**
N28	ControlArt	BL corner	122	**27.05**	10.48	0.95	38.8%	9.46	56.74	**43.75**
N31	ControlArt	BR corner	66	**18.88**	12.27	1.51	65.0%	5.67	62.18	**49.53**
N1	Ascott	TL corner	107	**19.87**	10.02	0.97	50.4%	6.85	69.08	**39.35**
N3	Ascott	TR corner	49	**23.19**	10.19	1.46	43.9%	7.60	54.54	**39.69**
N7	Ascott	Centre	62	**23.14**	7.82	0.99	33.8%	11.92	45.14	**35.98**
N11	Ascott	BL corner	59	**26.52**	8.46	1.10	31.9%	9.07	44.90	**41.46**
N13	Ascott	BR corner	114	**20.41**	8.52	0.80	41.7%	5.67	49.12	**34.95**

Table 13 shows that the two ControlArt centre panels N22 (36.23 µm) and N23 (42.58 µm) resulted in the highest means in the dataset, with p95 values of 71.50 and 71.41 µm, respectively. Secondly, the Ascott panels show systematically lower SD values (range 7.82–10.19 µm) than the ControlArt panels (range 10.48–19.21 µm), meaning that pit depths within any given Ascott panel were more similar to each other compared with the ControlArt panels. The Ascott centre panel N7 (CV% 33.8%) is the most internally consistent panel in the entire dataset.

Grouped Statistics and the Centre-to-Corner Gradient

Table 14: Grouped pit depth statistics for the zinc tiles…..

Group	n	Mean (µm)	SD (µm)	CV%	Max (µm)	p95 (µm)	Centre−corner gap (µm)
All panels — ControlArt	368	**27.85**	15.23	**54.7%**	93.75	**55.55**	—
Corners — ControlArt	299	24.88	12.53	50.3%	82.35	51.21	—
Centre — ControlArt	69	40.74	17.99	44.1%	93.75	71.41	**+15.9**
All panels — Ascott	391	**21.97**	9.31	**42.4%**	69.08	**38.91**	—
Corners — Ascott	329	21.75	9.56	43.9%	69.08	39.35	—
Centre — Ascott	62	23.14	7.82	33.8%	45.14	35.98	**+1.4**

Within the ControlArt chamber, the centre zinc tile panels experienced pits **64% deeper** than the corner panels (40.74 vs 24.88 µm). The p95 at the CA centre (71.41 µm) exceeded the CA corner p95 (51.21 µm) by 39%. This constitutes a pronounced and spatially systematic pattern. the centre of the ControlArt TC210 flat-bed chamber is a localised attack hotspot.

Within the Ascott chamber, no equivalent gradient could be observed. The centre mean (23.14 µm) exceeded the corner mean (21.75 µm) by only 1.4 µm. The Ascott chamber delivered spatially uniform pit depths across all positions.

Chamber Comparison and Mechanism Explanation

Table 12: Chamber comparison for pit depth of the zinc tiles…... Positive CA−AS values indicate that exposure in the ControlArt chamber resulted in deeper or more variable pits.

Metric	Ascott	ControlArt	CA−AS	Interpretation
Mean pit depth — all panels (µm)	**21.97**	**27.85**	**+5.88**	CA pits 27% deeper despite less total oxide. Controlled cycling concentrates attack at initiation sites.
SD pit depth — all panels (µm)	**9.31**	**15.23**	**+5.92**	CA wider spread — individual pit depths vary more.
CV% pit depth — all panels	**42.4%**	**54.7%**	**−12.3 pp**	AS more consistent overall. Lower CV% = more similar pit depths across the surface.
p95 pit depth — all panels (µm)	**38.91**	**55.55**	**+16.64**	CA worst-case pits 43% deeper. Most extreme pits substantially larger in CA.
Mean pit depth — corners (µm)	**21.75**	**24.88**	**+3.13**	CA corner pits deeper at 2 of 3 comparable corners.
CV% pit depth — corners	**43.9%**	**50.3%**	**−6.4 pp**	AS corner pits more consistent.
Mean pit depth — centre (µm)	**23.14**	**40.74**	**+17.60**	CA centre 76% deeper than AS centre. Most striking spatial difference in the dataset.
Centre vs corner gap (µm)	**≈+1.4**	**≈+15.9**	**—**	CA strong spatial gradient; AS essentially flat. CA centre is a localised attack hotspot.

Exposure in the ControlArt chamber resulted in deeper pits by every metric (+27% mean, +43% at p95), while exposure in the Ascott chamber resulted in more uniform pit depths (CV% 42.4% vs 54.7%). This is not a contradiction.

The **wet-dry transition** is the critical driver of localised pitting in cyclic corrosion testing. When the surface fully dries between spray cycles, the electrolyte concentrates at already-initiated pits. When it re-wets, those same sites are the first to reactivate, with a locally elevated chloride concentration. Each cycle deepens the same pits. The exposure conditions in the ControlArt chamber, completing the drying phase as specified, drove this repeated concentration-and-reactivation cycle in a strict way, resulting in deeper, more localised pits despite and less oxidation of total zinc.

The exposure in the Ascott chamber failed to complete the drying phase. As a result, the zinc surfaces were continuously or near-continuously wet. Without full de-wetting, the electrolyte could not concentrate at specific sites. Instead, corrosion was distributed broadly across the surface. The result was more corrosion product formation , since the zinc surface never dried out. **The two chambers were not operating at different intensities of the same mechanism. They were driving fundamentally different corrosion processes.**

This distinction matters for the interpretation of real component test results. On actual coated panels, pitting rather than uniform thinning is typically the damage mode responsible for paint delamination, creep, and eventual substrate perforation. A chamber that suppresses pitting and substitutes uniform dissolution may under-predict the formation of deep localised defects even while over-predicting total mass loss.

Key Takeaways and Chamber Assessment

A summary of key metrics from the zinc tile exposure to the ACT II cycle in the two chambers are presented in Table 16.

Table 13: Complete summary of key metrics

Metric	Ascott	ControlArt	Interpretation
CP: mean all tiles (g)	**0.02952**	**0.02009**	Ascott +47%. Extended wet phase from malfunction drove more total zinc dissolution.
CP: p95 all tiles (g)	**0.0612**	**0.0405**	Ascott +51% at worst case. Systematic excess across the full distribution.
CP: CV% all tiles	**55.6%**	**47.7%**	CA more consistent overall. Lower CV% = more uniform tile-to-tile exposure.
CP: CV% inner tiles	**56.7%**	**26.9%**	CA dramatically more consistent for inner tiles. Most significant homogeneity difference.
CP: CV% corners	**46.7%**	**72.9%**	AS more consistent at corners. CA corners highly non-uniform (N17 vs N28 differ ×6).
Pit depth: mean all (µm)	**21.97**	**27.85**	CA pits 27% deeper despite less oxide. Different corrosion mechanism, not just severity.
Pit depth: CV% all	**42.4%**	**54.7%**	AS more consistent. Pit depths more similar to each other across all panels.
Pit depth: p95 all (µm)	**38.91**	**55.55**	CA worst-case pits 43% deeper.

Pit depth: centre mean (µm)	**23.14**	**40.74**	CA centre 76% deeper than AS centre. Strong spatial hotspot in CA only.
Centre vs corner pit gap (µm)	**≈+1.4**	**≈+15.9**	CA strong gradient; AS flat. Specimen position matters significantly in CA.

The following findings emerge from the combined analysis:

1. The corrosive conditions of the Ascott chamber were systematically more severe.

The mean corrosion product mass was 47% higher after exposure in the Ascott chamber compared with corresponding exposure in the ControlArt chamber (the 95th percentile was 51% higher). This behaviour was consistent across all position groups (corners, centre, inner) and is directly explained by the extended wet phase caused by the temperature malfunction.

2. The corrosive conditions of the ControlArt chamber were more consistent for the corrosion product formation overall and particularly for the inner tile positions.

The CA inner tile CV% (26.9%) was less than half of AS inner tile CV% (56.7%). For most of the chamber floor area, the ControlArt chamber exposure resulted by far in a more uniform exposure. For the corner tiles, the Ascott chamber was more consistent (46.7% vs 72.9%), with the corners of the ControlArt's chamber showed non-uniform results driven by the large gap between N28 (BL) and N17 (TR).

3. The corrosive conditions of the ControlArt chamber resulted in deeper pits of the zinc tiles despite less total corrosion.

The CA mean pit depth exceeded findings in the Ascott chamber by 27% (27.85 vs 21.97 µm) and by 43% at the 95th percentile (55.55 vs 38.91 µm). Exposure in the Ascott chamber resulted in more uniform pit depths (CV% 42.4% vs 54.7%). This reflects a fundamental difference in corrosion mechanism with the ControlArt chambers complete wet-dry cycling concentrating the corrosion process locally, and the Ascott's chambers incomplete drying spreading the corrosion processuniformly.

4. The corrosive conditions of the ControlArt chamber resulted in a pronounced spatial gradient for localised corrosion – an effect not observed in the Ascott chamber.

The CA centre panels showed pits 64% deeper than the CA corner panels. In the Ascott chamber, the zinc tiles in the centre and at the corners showed essentially identical pit depths (6% difference). The centre of the ControlArt TC210 flat-bed chamber is evidently more corrosive hotspot, which has direct implications for specimen placement in future tests.

5. Suitability for future testing differs by objective.

Both chambers showed the same results on the qualitative spatial patterns and relative severity ordering. For *qualification and certification testing*, where absolute exposure severity must conform to the ACT II specification, the Ascott chamber is in its current state not suitable. A +47% mean higher formation of corrosion products and a suppressed pitting mechanism represent systematic biases that could result in false positive outcomes (specimens failing a criterion they would pass under correctly controlled conditions). It would also fail to correctly reproduce the localised corrosion process mode of relevance to real component degradation. Repair and re-validation of the Ascott chamber are required before it can be used for qualification testing.

5.3 WP3: Scribed Panels

Descriptive statistics

Table 17 presents the global descriptive statistics for each material, computed across all conditions, using position averages as the unit of analysis (n = 12 per material). The statistics reported include arithmetic mean,

standard deviation (SD), standard error (SE = SD/√n), minimum, maximum, median, and coefficient of variation (CV% = SD/mean × 100).

Table 14: Overall descriptive statistics by material (n = 12 position averages per material, across both chambers and both orientations). * CV% for A2202F03 is not interpretable due to near-zero mean

Material	n	Mean (mm)	SD (mm)	SE (mm)	Min (mm)	Max (mm)	Median (mm)	CV%
A2202F03 Aluminium	12	**0.025**	0.062	0.018	0.000	0.200	0.000	—*
ZM2406F08 Zn-Mg steel	12	**1.762**	0.379	0.109	1.338	2.792	1.681	21.5%
H2507F08 Galvanized	12	**3.484**	0.748	0.216	2.752	5.176	3.342	21.5%
S2304F10 Bare steel	12	**4.230**	0.354	0.102	3.732	4.824	4.176	8.4%

The results confirm a clear increasing gradient of corrosion severity across the four materials. The aluminium alloy resulted in a mean propagation of just 0.025 mm with a median of 0.000 mm. The single non-zero position average arose from one isolated segment measurement (Segment IV, 1.00 mm in the ControlArt diagonal condition at position 2), which averaged to 0.200 mm across the five segments.

ZM2406F08 showed moderate corrosion (mean 1.762 mm), **H2507F08** showed significant propagation (mean 3.484 mm), and **S2304F10** showed the highest and most consistent corrosion (mean 4.230 mm, CV% = 8.4%). Notably, S2304F10 had the lowest CV% of all materials with meaningful corrosion, reflecting the fact that bare steel corrodes severely and uniformly under any condition that includes chloride deposition.

Subgroup Breakdown by Material, Chamber and Orientation

Table 18 presents the complete descriptive statistics for each combination of material, chamber and scribe orientation (n = 3 per subgroup). This table is the primary data reference for all subsequent analyses.

Table 18: Descriptive statistics by material × chamber × orientation (n = 3 position averages per subgroup).

Material	Chamber	Orientation	n	Mean (mm)	SD (mm)	Min (mm)	Max (mm)	CV%
A2202F03	ControlArt	Diagonal	3	**0.067**	0.115	0.000	0.200	173%
A2202F03	ControlArt	Horizontal	3	**0.033**	0.058	0.000	0.100	173%
A2202F03	Ascott	Diagonal	3	**0.000**	0.000	0.000	0.000	—
A2202F03	Ascott	Horizontal	3	**0.000**	0.000	0.000	0.000	—
ZM2406F08	ControlArt	Diagonal	3	**1.610**	0.144	1.448	1.722	8.9%
ZM2406F08	ControlArt	Horizontal	3	**2.249**	0.471	1.948	2.792	20.9%
ZM2406F08	Ascott	Diagonal	3	**1.473**	0.118	1.338	1.550	8.0%
ZM2406F08	Ascott	Horizontal	3	**1.715**	0.116	1.606	1.836	6.7%
H2507F08	ControlArt	Diagonal	3	**2.880**	0.120	2.752	2.990	4.2%
H2507F08	ControlArt	Horizontal	3	**3.282**	0.182	3.096	3.460	5.5%
H2507F08	Ascott	Diagonal	3	**3.689**	0.948	2.870	4.728	25.7%
H2507F08	Ascott	Horizontal	3	**4.085**	0.956	3.394	5.176	23.4%
S2304F10	ControlArt	Diagonal	3	**4.250**	0.310	3.908	4.512	7.3%
S2304F10	ControlArt	Horizontal	3	**4.204**	0.353	3.908	4.594	8.4%
S2304F10	Ascott	Diagonal	3	**3.888**	0.154	3.732	4.040	4.0%
S2304F10	Ascott	Horizontal	3	**4.576**	0.300	4.242	4.824	6.6%

Several patterns in Table 2 warrant attention. For **H2507F08** exposed in the Ascott chamber, both the diagonal (CV = 25.7%) and horizontal (CV = 23.4%) subgroups show large standard deviations (0.948 and 0.956 mm respectively), indicating substantial position-to-position scatter within the chamber for this material. In contrast, the same material exposed in the ControlArt chamber shows SDs of just 0.120 and 0.182 mm (CV = 4.2% and 5.5%).

For **ZM2406F08** in ControlArt, the horizontal orientation shows a larger SD (0.471 mm, CV = 20.9%) compared to the diagonal (0.144 mm, CV = 8.9%), driven by position 2 recording a mean of 2.792 mm against 1.948 mm and 2.008 mm at the other two positions.

Scribe Orientation Effect

Table 19 compares the mean corrosion propagation for diagonal and horizontal scribes within each material × chamber combination, reporting the H−D mean difference. A positive value indicates that the horizontal scribe produced greater corrosion propagation.

Table 15: **Orientation effect of the scribed materials exposed….. : mean propagation for diagonal (D) and horizontal (H) scribes, with H−D difference. Positive H−D values indicate horizontal scribe was more severely corroded.**

Material	Chamber	D mean (mm)	D SD	H mean(mm)	H SD	H−D (mm)	Observation
A2202F03	ControlArt	0.067	0.115	**0.033**	0.058	**−0.033**	Negligible, near-zero in both. No meaningful effect.
A2202F03	Ascott	0.000	0.000	**0.000**	0.000	**0.000**	All zero. Not applicable.
ZM2406F08	ControlArt	1.610	0.144	**2.249**	0.471	**+0.639**	Strong H effect (+40%). H pos-2 (2.79mm) amplifies gap.
ZM2406F08	Ascott	1.473	0.118	**1.715**	0.116	**+0.241**	Moderate, H effect (+16%). Consistent across all positions.
H2507F08	ControlArt	2.880	0.120	**3.282**	0.182	**+0.402**	Clear H effect (+14%). Consistent across all positions.
H2507F08	Ascott	3.689	0.948	**4.085**	0.956	**+0.396**	Same absolute gap (+11%) despite higher overall scatter.
S2304F10	ControlArt	4.250	0.310	**4.204**	0.353	**−0.046**	No effect (−1%). Both

							directions near-saturated.
S2304F10	Ascott	3.888	0.154	**4.576**	0.300	**+0.688**	H effect (+18%). Can be due to malfunction

The horizontal scribe resulted in greater corrosion propagation than the diagonal in six of the eight material × chamber combinations. The physical mechanism was consistent across all cases. During the ACT II salt rain phases, the horizontal scribe geometry acts as a trough that retains the electrolyte solution against the exposed material surface for longer than the diagonal scribe, which allows the solution to drain by gravity during transition and drying phases. This increases the cumulative chloride loading and extends the time of wetness at the scribe interface, both primary kinetic drivers of atmospheric corrosion.

For **ZM2406F08 in ControlArt**, the horizontal mean was 2.249 mm against 1.610 mm diagonal, an H−D difference of +0.639 mm (+40%). This is the largest relative orientation effect in the dataset, though it is partly amplified by the outlying position 2 horizontal value (2.792 mm). For **ZM2406F08 in Ascott**, the H−D gap narrows to +0.241 mm (+16%), and is consistent across all three positions, suggesting a genuine but smaller orientation effect.

For **H2507F08**, the absolute H−D gap is +0.402 mm upon exposure in the ControlArt chamber and +0.396 mm in the Ascott chamber, essentially identical in both chambers despite the very different overall corrosion levels. This is an important finding as the orientation effect for galvanized steel is *chamber-independent*, indicating that it reflects a material response to electrolyte retention geometry rather than a chamber-specific behaviour.

For **S2304F10**, no orientation effect was observed upon exposure in thee ControlArt chamber (−0.046 mm, −1%), as both scribe directions had already reached near-maximum corrosion. In the Ascott chamber, an unexpected H effect of +0.688 mm (+18%) was observed, most probably being an artefact of the temperature malfunction extending the effective wet time at the horizontal scribe rather than a genuine material response to orientation.

For **A2202F03**, the near-zero corrosion in all conditions renders any orientation comparison physically meaningless.

Variability Analysis: Coefficient of Variation

The coefficient of variation (CV%) measures dispersion relative to the mean, providing a scale-free index of consistency:

CV% = (Standard Deviation / Mean) × 100

A low CV% indicates that all three panel positions within a given exposure condition resulted in similar corrosion propagation, meaning that the chamber delivered spatially uniform exposure. A high CV% indicates large position-to-position scatter, suggesting that the exposure conditions varied significantly across the chamber volume. Thresholds applied: CV% < 10% = very consistent; 10–25% = moderate variability; > 25% = high variability.

Table 16: Coefficient of variation (CV%) by material × chamber × orientation (n = 3 per subgroup). * CV% not interpretable for A2202F03 due to near-zero mean.

Material	Chamber & Orientation	Mean (mm)	SD (mm)	CV%	n	Interpretation

A2202F03	ControlArt Diagonal	**0.067**	0.115	*	1	Near-zero mean. CV not meaningful.
A2202F03	ControlArt Horizontal	**0.033**	0.058	*	3	Near-zero mean. CV not meaningful.
A2202F03	Ascott Diagonal	**0.000**	0.000	**undef**	3	All zero. CV undefined.
A2202F03	Ascott Horizontal	**0.000**	0.000	**undef**	3	All zero. CV undefined.
ZM2406F08	ControlArt Diagonal	**1.610**	0.144	**8.9%**	3	Very consistent diagonal exposure in ControlArt.
ZM2406F08	ControlArt Horizontal	**2.249**	0.471	**20.9%**	3	Moderate. position 2 (2.79 mm) drives the scatter.
ZM2406F08	Ascott Diagonal	**1.473**	0.118	**8.0%**	3	Very consistent. Ascott diagonal uniform for ZM.
ZM2406F08	Ascott Horizontal	**1.715**	0.116	**6.7%**	3	Very consistent. Ascott horizontal also uniform for ZM.
H2507F08	ControlArt Diagonal	**2.880**	0.120	**4.2%**	3	Excellent. all 3 positions within 0.24 mm of each other.
H2507F08	ControlArt Horizontal	**3.282**	0.182	**5.5%**	3	Good. well-controlled drying phase in ControlArt.
H2507F08	Ascott Diagonal	**3.689**	0.948	**25.7%**	3	HIGH. position 2 (4.73 mm) vs positions 1 & 3 (~2.87–3.47 mm). 1.86 mm spread.
H2507F08	Ascott Horizontal	**4.085**	0.956	**23.4%**	3	HIGH. position 2 (5.18 mm) vs positions 1 & 3 (~3.39–3.69 mm). 1.78 mm spread.
S2304F10	ControlArt Diagonal	**4.250**	0.310	**7.3%**	3	Very consistent. severe corrosion uniform across positions.
S2304F10	ControlArt Horizontal	**4.204**	0.353	**8.4%**	3	Very consistent.
S2304F10	Ascott Diagonal	**3.888**	0.154	**4.0%**	3	Very consistent.
S2304F10	Ascott Horizontal	**4.576**	0.300	**6.6%**	3	Low-moderate. position 3 (4.82 mm) slightly higher.

One of the most significant findings of these samples tests is that the CV% analysis revealed a material-dependent reversal of chamber consistency. . For **H2507F08**, exposure in the ControlArt chamber was far more consistent (CV 4.2–5.5%) than corresponding exposure in the Ascott chamber (CV 23.4–25.7%). For **ZM2406F08**, the pattern was reverse: the Ascott chamber exposure was more consistent (CV 6.7–8.0%) than in the ControlArt chamber (CV 8.9–20.9%). This reversal rules out a simple 'one chamber is generally better controlled' explanation and instead points to material-specific sensitivities to different aspects of the exposure cycle.

The high CV% values observed for H2507F08 in the Ascott chamber are driven almost entirely by position 2. The individual segment readings at diagonal position 2 were [5.44, 4.68, 4.63, 4.74, 4.15] mm. All five segments consistently elevated, confirming this is a genuine panel-level exposure difference rather than a

measurement artefact. The panel at position 2 in experienced demonstrably more severe corrosion than panels exposed at positions 1 and 3 under nominally identical conditions. This is the clearest quantitative evidence in the dataset of spatial non-uniformity within the Ascott chamber, consistent with the documented temperature malfunction creating locally extended wet phases at that position.

The elevated CV% for ZM2406F08 in the ControlArt chamber is attributable to a single outlying horizontal position (position 2: 2.792 mm versus 1.948 and 2.008 mm at the other two positions). Without this position, the horizontal CV% would fall below 5%. This may represent a test-specific artefact, which cannot be determined without replication.

For **S2304F10**, CV% values are low in all conditions (3.9–10.2%), confirming that bare steel corrodes consistently and severely regardless of chamber or orientation. For **A2202F03**, the CV% is not interpretable due to the near-zero mean and the absence of meaningful corrosion propagation.

Chamber Comparison

Table 21 presents the mean corrosion propagation, standard deviation, and CV% for each material in each chamber, pooling both scribe orientations (n = 6 position averages per group). The final column shows the CA−AS difference (positive = ControlArt produced more corrosion; negative = Ascott produced more).

The Ascott chamber temperature malfunction must be kept in mind throughout this section.

Table 17: Chamber comparison of the ….. : mean, SD, and CV% for ControlArt (CA) and Ascott (AS) per material (n = 6, both orientations pooled). * CV% for A2202F03 not interpretable.

Material	CA mean	CA SD	CA CV%	AS mean	AS SD	AS CV%	CA − AS
A2202F03 Aluminium	**0.050**	0.084	167%*	**0.000**	0.000	—	**+0.050**
ZM2406F08 – Zn-Mg steel	**1.930**	0.469	24.3%	**1.594**	0.168	10.6%	**+0.336**
H2507F08 Galvanized	**3.081**	0.260	8.4%	**3.887**	0.879	22.6%	**−0.806**
S2304F10 – Bare steel	**4.227**	0.298	7.0%	**4.232**	0.433	10.2%	**−0.005**

Materials With No Chamber Effect

Exposure of **A2202F03** resulted in both chambers in minor or no corrosion (CA mean: 0.050 mm; AS mean: 0.000 mm). The difference of 0.050 mm is attributable to two isolated non-zero segment readings for the ControlArt chamber exposure. The aluminium oxide self-passivation mechanism suppresses corrosion propagation under any condition, making this material insensitive to chamber differences.

For **S2304F10**, the chamber means differed by only 0.005 mm (CA: 4.227 mm; AS: 4.232 mm). Exposure in both chambers resulted in statistically and practically identical results. The bare steel corroded at or near its maximum rate under any chloride-containing cyclic exposure regardless of the specific temperature profile or chamber geometry, leaving no room for any chamber differences to be discerned.

ZM2406F08

Exposure in the ControlArt chamber resulted in a mean of 1.930 mm for ZM2406F08 versus 1.594 mm in the Ascott chamber, a difference of +0.336 mm. This direction is counter intuitive. If the temperature malfunction of the Ascott chamber had made its exposure conditions more severe, it would be expected to result in more corrosion products as it was “over-exposing” the specimens. The more likely explanation relates to the

sensitivity of the zinc-magnesium coating to the *peak drying-phase temperature*. If the temperature in the ControlArt chamber consistently reached 50°C during the drying phase, while the malfunctioning Ascott chamber operated at lower effective temperatures, the difference in peak temperature could drive different corrosion product phase formation kinetics at the ZM scribe interface, resulting in slightly more lateral corrosion in the ControlArt chamber.

From a consistency perspective, the exposure conditions of the Ascott chamber compared with the ControlArt chamber resulted in a better performance of ZM2406F08 (CV 10.6% vs ControlArt 24.3%). The Ascott position averages for this material fell within a 0.498 mm band; the ControlArt averages spanned 1.344 mm. As noted previously, the scatter in the ControlArt chamber is mainly attributed to one of the outlying horizontal positions (which with repeating this test would translate into a better repeatability of ControlArt chamber).

H2507F08

Exposure in the Ascott chamber resulted in a mean of 3.887 mm for H2507F08 compared with 3.081 mm in the ControlArt chamber, a difference of −0.806 mm. This is the largest chamber-associated mean difference in the dataset. The Ascott standard deviation (0.879 mm, CV = 22.6%) was more than three times larger than ControlArt's (0.260 mm, CV = 8.4%).

Table 18: Individual position averages (mm) for all materials in both chambers (D pos 1–3 = diagonal positions; H pos 1–3 = horizontal positions). * marks outlying positions driving chamber differences.

Material	Chamber	D pos 1	D pos 2	D pos 3	H pos 1	H pos 2	H pos 3
A2202F03	ControlArt	0.000	0.200	0.000	0.000	0.000	0.100
A2202F03	Ascott	0.000	0.000	0.000	0.000	0.000	0.000
ZM2406F08	ControlArt	1.722	1.448	1.660	1.948	*** 2.792**	2.008
ZM2406F08	Ascott	1.338	1.532	1.550	1.702	1.606	1.836
H2507F08	ControlArt	2.752	2.990	2.898	3.290	3.096	3.460
H2507F08	Ascott	2.870	*** 4.728**	3.470	3.686	*** 5.176**	3.394
S2304F10	ControlArt	3.908	4.512	4.330	4.110	3.908	4.594
S2304F10	Ascott	3.892	3.732	4.040	4.242	4.662	4.824

For **H2507F08** exposed in the ControlArt chamber, all six position averages fall within a tight 0.71 mm band (2.752 to 3.460 mm). For H2507F08 in the Ascott chamber, five of six positions cluster between 2.870 and 3.686 mm, broadly consistent with findings in the ControlArt chamber, but **position 2** in both orientations records values of 4.728 mm (diagonal) and 5.176 mm (horizontal), approximately 1.3–1.5 mm above all other positions in the same chamber. These two extreme values alone account for the higher t mean and virtually all of the higher CV% observed for the materials exposed in the Ascott chamber.

The most coherent explanation is that the panel at position 2 in the Ascott chamber experienced a locally extended wet phase as a consequence of the temperature malfunction. If the chamber failed to reach 50 °C consistently in the region occupied by position 2, the drying phase would have been less effective at removing the electrolyte from the scribe channel, leaving the galvanized surface in prolonged contact with the chloride-containing solution. This would accelerate anodic dissolution of the zinc coating and increase the lateral propagation well beyond what would occur at real conditions.

In practical terms, the exposure conditions in the ControlArt chamber provided a more *reproducible* test environment for H2507F08, with all panels producing results consistent with the expected galvanic protection behaviour of the zinc coating. Parallel exposure in the Ascott chamber resulted in five of six positions that were

broadly comparable to the findings of the ControlArt exposure, but the anomalous position 2 undermines the reliability of the overall result for this material in the Ascott chamber.

Summary and Conclusions

The final material ranking in terms of mean corrosion propagation from the scribe is presented in Table 23.

Table 19: Final material ranking by overall mean corrosion propagation (n = 12 per material, both chambers and orientations combined).

Rank	Material	Mean (mm)	Summary	Grade
1	**A2202F03 Aluminium**	**0.025**	Near-zero in all conditions. Aluminium secondary oxide film arrests corrosion propagation at the scribe. Chamber and orientation irrelevant.	**EXCELLENT**
2	**ZM2406F08 Zn-Mg steel**	**1.762**	Moderate propagation. Zn-Mg coating provides galvanic + barrier protection. Sensitive to horizontal scribe orientation, especially in ControlArt (+0.64 mm).	**GOOD**
3	**H2507F08 Galvanized**	**3.484**	Significant propagation. Galvanic protection partially effective. Most condition-sensitive material — high scatter in Ascott (CV 22.6%) vs very consistent in ControlArt (CV 8.4%).	**POOR**
4	**S2304F10 Bare steel**	**4.230**	Severe and consistent corrosion. No protective system; substrate corrodes maximally in all conditions. Chamber and orientation effects negligible.	**WORST**

The following conclusions are drawn from the complete analysis:

The Material is the primary determinant of corrosion propagation.

The four materials are ranked unambiguously from best to worst (A2202F03 < ZM2406F08 < H2507F08 < S2304F10). This ranking is stable across both chambers, both scribe orientations, and all panel positions.

Horizontal scribes generally results in more corrosion than diagonal scribes.

The effect is consistent for ZM2406F08 and H2507F08. For H2507F08, the H−D gap is almost identical in both chambers (+0.40 mm), confirming it is a material response to the electrolyte retention geometry rather than a chamber effect. The physical mechanism is the trough-like retention of the salt solution in the horizontal scribe channel during ACT II wet phases.

The exposure conditions of the ControlArt chamber resulted in more reproducible results for H2507F08 compared to the Ascott chamber.

Exposure in the ControlArt chamber resulted in a CV% of 4.2–5.5% versus 23.4–25.7% in the Ascott chamber. All six ControlArt positions fell within a 0.71 mm band; Ascott position 2 (diagonal: 4.73 mm, horizontal: 5.18 mm) was approximately 1.3–1.5 mm above the other five Ascott positions. The anomaly at position 2 in the Ascott chamber is consistent with the temperature malfunction creating locally extended wet phases at that location.

The exposure conditions of the Ascott chamber resulted in more reproducible exposure for ZM2406F08 compared to the ControlArt chamber.

Exposure in the Ascott chamber resulted in a CV% of 6.7–8.0% i versus 8.9–20.9% in the ControlArt chamber. The scatter in the latter was driven by one outlying horizontal position. For this material, exposure in the Ascott chamber resulted in more spatially uniform conditions. It is likely that if repeated, exposure in the ControlArt chamber will be more reproducible, since if that scatter is not considered, the CV% would be below 5%.

The chamber effect reverses the results obtained in the two chambers between the two sensitive materials.

Exposure in the ControlArt chamber resulted in a higher mean for ZM2406F08 (+0.336 mm), while the exposure in the Ascott chamber resulted in a higher mean for H2507F08 (+0.806 mm). None of the chambers are simply 'more severe' from a corrosion perspective. The two coatings appear to be sensitive to different aspects of the exposure cycle. Exposure parameters such as peak drying temperature for ZM2406F08, and wet phase duration and continuity for H2507F08 were differently delivered by the two chambers.

A confirmed repeat test is required for definitive chamber conclusions.

The Ascott chamber temperature malfunction introduces an uncontrolled confounding variable of unknown magnitude. A repeat test with both chambers verified to maintain ACT II setpoints throughout is needed to isolate genuine chamber design differences from malfunction-driven artefacts.

5.4 WP4: Crevice Panels

Overview of Post-Exposure Appearance



Overview photograph of all Ascott crevice specimens after 6 weeks

Overview photograph of all ControlArt TC210 crevice specimens after 6 weeks

Figure 16 Test crevice panels exposed in the Ascottand ControlArt chambers for 6 weeks of the ACT II Cycle

Most visually striking corrosion was observed for the TF specimens from both chambers. The inner surface of the main coupon displayed substantial, voluminous orange-brown iron oxide deposits covering the full width of the overlap area. For the Ascott chamber-exposed specimens, the oxide layer was thick, continuous, and showed a laminated texture consistent with repeated wet-dry cycling, with multiple layers of corrosion product accumulated over the six-week exposure. The deposit was thickest in the centre of the overlap, tapering slightly toward the lateral edges where drainage during the drying phase was more effective.

The TF specimens exposed in the ControlArt chamber showed qualitatively comparable, but *slightly different spatial distribution* of corrosion products within the crevice zone. The oxide deposit was again substantial and continuous but showed a more pronounced **central concentration** which would be expected in a chamber with more effective drying compared to a chamber with incomplete drying (the Ascott chamber temperature malfunction).

Phosphate (PH) specimens

The phosphate-treated specimens showed consistently milder corrosion in the spacer-crevice zone compared to their TF counterparts. For the PH specimens exposed in the Ascott chamber, corrosion products were concentrated as a narrow band along the length of the crevice, thinner and less voluminous than in the TF group. The paint at the crevice mouth remained partially adherent, and only local delamination was observed. The surface of the main coupon retained visible zinc coating in the areas furthest from the crevice mouth, indicating that the galvanising was still providing cathodic protection to portions of the interface.

For the PH specimens exposed in the ControlArt chamber, the crevice-zone corrosion was notably mild. The most representative specimen, PH45, shows only a thin, discontinuous rust stain along one edge of the crevice face, with the remainder of the inner surface largely intact and retaining its zinc appearance. The zinc phosphate coating, combined with the partial paint penetration at the crevice mouth and the more effective drying in the ControlArt chamber, appears to have been sufficient to substantially limit the crevice corrosion initiation over the six weeks. By contrast, the PH specimens exposed in the Ascott chamber showed more developed corrosion under identical nominal conditions, consistent with the extended wet-phase exposure produced by the incomplete drying in the Ascott chamber.

CT Scan Assessment

The central crevice specimens from both chambers were scanned by X-ray CT prior to disassembly. The CT images were reviewed for evidence of sub-surface pitting at both the spacer-crevice interface and the joint-end interface, and for the three-dimensional distribution of corrosion product within the assembly.

The CT scan did **not reveal appreciable pitting** in any specimen from either chamber, figure 17. The corrosion product accumulated at the crevice interface was detectable as a high-density phase, confirming that corrosion products had formed within the crevice volume, but no sub-surface pits extending into the base metal were resolved.

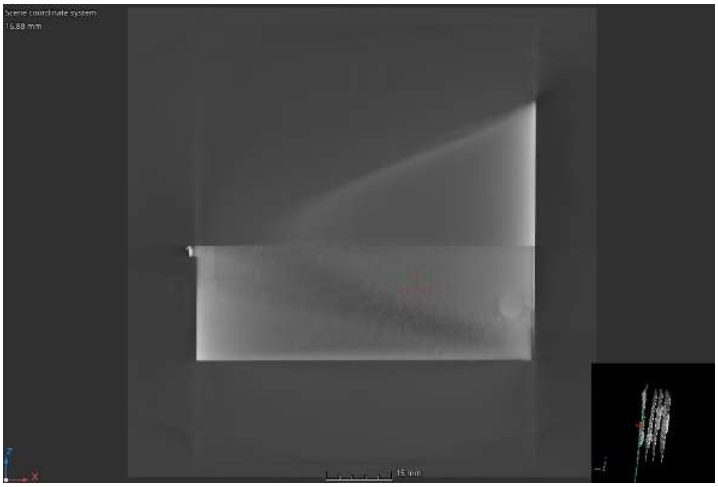


Figure 17 Scan of a thin sample exposed to ACTII

5.5 WP5: Complex Components

Cross-sectional metallographic analysis was performed on one specimen per exposure chamber, selected as the most visually corroded representative of each exposure condition. While the limited sample size precludes statistical inference, the examination provided a qualitative indication of corrosion depth and morphology in each chamber serves as a baseline for more systematic future work.

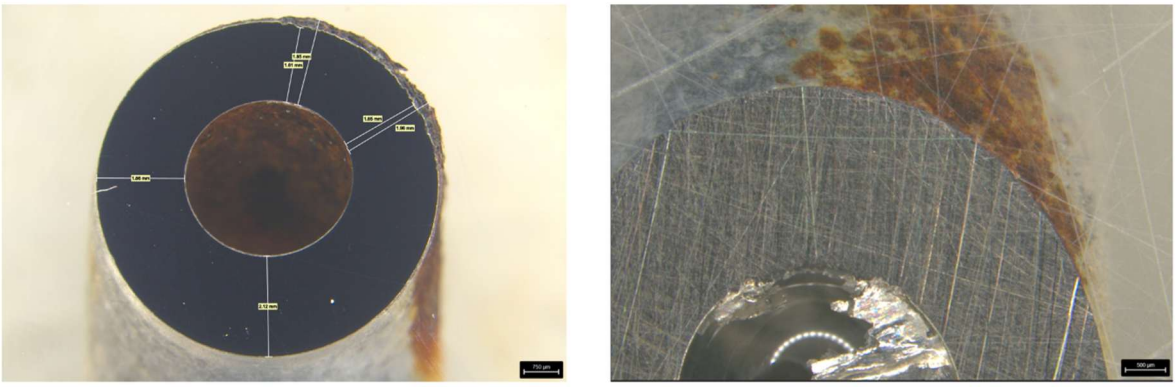


Figure 18: Brake hose Cross cut comparison

For the ControlArt chamber exposed specimen, optical microscopy of the polished cross-section revealed a well-defined corrosion layer extending inward from the outer surface of the component, figure 17. Thickness measurements taken at multiple angular positions around the section yielded values of 1.85 mm, 1.61 mm, 1.65 mm and 1.96 mm, indicating moderately non-uniform corrosion. The corrosion front was clearly distinguishable from the underlying base material, and the layered structure of the corrosion products was visible within the corroded zone. The outer edge of the section also showed macroscopically visible rust, consistent with the progressive surface corrosion documented during weekly inspections.

In contrast, the specimen exposed in the Ascott chamber did not exhibit a comparably developed corrosion layer in cross-section. Surface examination under the optical microscope revealed the presence of rust products at the outer surface, consistent with the corrosive environment, but no measurable penetration depth analogous to that observed in the specimen exposed in the ControlArt chamber could be identified. Corrosion for this specimen appeared to remain largely superficial over the six-week exposure period.

The difference in corrosion depth between the two chambers suggests that the ControlArt cycle promoted more severe and deeper material degradation in steel and ZiNi coated specimens over the same exposure duration. However, given that only one specimen per chamber was analysed, this observation should be interpreted with caution. Variability in specimen geometry, local surface condition, and positioning within the chamber could all influence the results. A more robust comparison, involving a larger number of specimens and replicated cross-sections at multiple locations, is planned as part of future work to determine whether this difference is reproducible and to what extent it can be attributed to the differing test cycle parameters.

6 ACT II suggested Improvement

Mandatory standardised calibration programme

In the current standard, a prescribed calibration test programme is missing. VCS 1027,1449 specifies what the chamber must achieve in terms of mean values and maximum fluctuation, but not how to verify it. Different

users may hence verify compliance by monitoring conditions during a live ACT II run, by running a shortened subset of setpoints, or by relying on the chamber manufacturer's factory calibration certificate. None of these approaches is equivalent, and none guarantees that all three required setpoints have been held for a sufficient duration under controlled, reproducible conditions.

The standard should define a mandatory calibration programme consisting of three discrete steps, one per setpoint (25 °C/95% RH, 50 °C/70% RH, and 50 °C/95% RH), each held for a minimum of five hours. The five-hour duration provides sufficient data for a reliable statistical evaluation of both mean accuracy and short-term fluctuation. The first sixty minutes of each step should be explicitly excluded from the statistical evaluation to account for ramp transients and thermal stabilisation. The total programme duration of fifteen hours is operationally realistic and can be completed overnight without interrupting production testing schedules.

No spray during calibration

A requirement that is currently implicit but never explicitly stated in the standard is that the calibration programme must be conducted with the salt spray, or salt rain system inactive. This distinction is critical. When the spray system is operating, evaporation of the salt solution at the specimen surfaces introduces an uncontrolled and position-dependent source of humidity input into the chamber atmosphere. The latent heat of evaporation simultaneously perturbs the temperature field. Consequently, the temperature and humidity readings obtained during a live ACT II run cannot be attributed solely to the chamber's climate control system.

By running the calibration programme in dry mode (climate only, spray inactive), the measurements isolate the chamber's intrinsic ability to regulate temperature and relative humidity at each setpoint. This is precisely the quantity that the DAS and MSPDA limits are intended to assess. If calibration were permitted with spray active, a chamber with poor relative humidity regulation could pass the MSPDA criterion based on spray-driven humidification rather than genuine humidity control.

Probe number, type and positioning

The current standard does not specify how many independent probes must be used during calibration, where they should be positioned within the test volume, or what measurement uncertainty is acceptable. This study showed that in both the Ascott and the ControlArt chambers, the internal sensors, and the free-air external probes resulted in measurably different readings, with systematic differences in relative humidity at the canopy position exceeding the DAS limit in both chambers. Relying on the internal sensor alone would have given a falsely optimistic assessment of both chambers.

The standard should require a minimum of two independent external probes in addition to the internal sensor, positioned at defined locations: one at the geometric centre of the test area at the test area height, and one at the upper region of the chamber near the canopy, where the thermal insulation is most likely to produce measurement gradients. Both probes should be mounted vertically on a specimen support to prevent water accumulation on the sensor element. Probe measurement uncertainty should not exceed ±0.2 °C for temperature and ±1.5% RH for humidity, verified by traceable calibration. The DAS and MSPDA acceptance limits should be applied to each sensor independently, so that spatial non-uniformity is explicitly captured in the pass/fail outcome rather than masked by averaging across sensors.

Standardised Chamber Loading Condition During Calibration

A gap that is rarely discussed in the context of corrosion chamber calibration is the loading state of the chamber at the time of verification. The thermal mass, surface area, and moisture absorption characteristics of the

specimens loaded into a chamber during a production test can measurably alter the temperature and relative humidity dynamics compared to an empty chamber. A chamber calibrated empty and then loaded to full capacity with metal panels and component assemblies will not necessarily exhibit the same climate control performance that was verified during calibration. The reason is that the specimens absorb and release heat and moisture in ways that interact with the PID control loop.

This is a complex parameter to standardise since loading conditions can be considered by either mass or volume. For example, zinc foils occupy a very different volume and mass to a set of full body panels or brake assemblies. The standard should address this by requiring an empty chamber during calibration testing so that no extra requirements and verification on quantity of sample inside a chamber is necessary.

7 Research questions answered

The research questions raised at the start of this master thesis study and their answers are given below.

RQ1 Can a comparative methodology using a standardised set of samples be developed to determine whether a new chamber can follow the Volvo standard corrosion cycles?

Answer: Yes.

A structured five-layer methodology was successfully developed and validated. It progresses from environmental characterisation (WP1) through spatial aggressiveness mapping with zinc foils (WP2), scribed panel corrosion propagation (WP3), crevice corrosion testing (WP4), and real automotive sub-assemblies (WP5). Each layer adds a dimension of evidence that the previous one cannot provide: WP1 identifies whether the chamber meets the environmental specification; WP2 reveals spatial non-uniformities in corrosive severity; WP3–WP5 establish whether those differences translate into measurably different outcomes at the material and component level. The workflow is hardware-agnostic and reusable.

RQ2 Given the Ascott chamber temperature malfunction, do the results fall within a range where they remain comparable with a functioning ACT II chamber (ControlArt)?

Answer: Partially, comparability depends on what is being measured.

The Ascott chamber failure to consistently reach the 50 °C drying-phase setpoint introduced a systematic bias throughout the campaign. Quantitatively, the results are not comparable: zinc foil inner tiles showed 69% higher corrosion mass loss in the Ascott chamber compared with the ControlArt chamber (0.033 g vs 0.019 g), a difference directly attributable to extended time of wetness during the incomplete drying phase. For quantitative pass/fail assessments, the two chambers cannot be treated as equivalent without applying a correction. Qualitatively, however, material rankings and relative damage distributions were largely preserved across WP3–WP5, meaning that the results obtained using the Ascott chamber retain value for material selection decisions, even in the presence of the malfunction.

RQ3 Scope for Standard Improvement: Which aspects of the current standard's calibration and acceptance criteria lack sufficient specificity, and what modifications or stricter requirements could help reduce inter-laboratory variability?

Answer: Five possible improvements were identified, with one of overriding priority.

The most significant improvement that can be implemented is adding a standardised calibration programme. The standard specifies acceptance limits (DAS, MSPDA) but not how to verify them, leaving the verification procedure open to interpretation and making calibration outcomes from different laboratories

fundamentally incomparable. The solution is a prescribed three-step dry-mode programme (one step per setpoint, five hours each, first sixty minutes excluded, spray inactive). Running calibration without spray is essential. When the spray system is active, evaporation perturbs both temperature and humidity, making it impossible to evaluate the chamber's intrinsic climate control independently of the spray contribution.

Other possible improvements include the definition of minimum probe number and positioning, cooling time verification against the centre probe, and empty chamber condition (very difficult to define how full a chamber should be). Volume high is different than mass high. Different implication, high in volume means for example less airflow. Taken together, these improvements can result in a systematic expansion of the standard from a set of acceptance limits into a fully specified, reproducible verification procedure.

8 Use of generative AI

Generative artificial intelligence tools were used in this research to support the writing and the coding process (debugging and plotting). Claude was used for grammar correction and clarity of expression to keep an academically and appropriate language. No technical, methodological, analysis and interpretations were generated directly by AI, but it was an independent work brought out by the author. AI suggestions were critically reviewed, edited, and validated to ensure full academic integrity and alignment with the intended scope of the master thesis project.

9 Future work

9.1 Future work

This thesis developed and validated a five-layer comparative methodology for accelerated corrosion chamber qualification, answered three core research questions, and proposed targeted improvements to the ACT II standard. Future sample testing, which is already ongoing is presented here as well with some suggestions for future retesting of the same materials as investigated in this master thesis study.

Rivet joints on different aluminium alloys

A natural and high-priority extension of WP5 is the testing of riveted joints on different aluminium alloy substrates. Riveted aluminium joints represent one of the most corrosion-critical assembly configurations in modern automotive body structures, combining galvanic coupling between the rivet and the sheet, crevice geometry at the lap joint, and residual stress around the rivet hole. Testing should include at least two aluminium alloy systems of different corrosion resistance. Grounding cable and connections are also to be tested along with brakes hose of different suppliers and with different surface treatments.

Zinc flake coating as a protective treatment

Zinc flake coatings applied by spray are widely used in automotive fastener and underbody protection applications as an alternative to electrodeposited zinc. Their performance under ACT II conditions in the context of chamber comparability has not been studied. Zinc flake coatings are currently being tested for their protection mechanism that depends critically on the availability of zinc dissolution products to form barrier layers.

ACT III and walk-in chambers

The master thesis explicitly identified the extension to "walk-in" chambers and the ACT III cycle for large automotive components as future work. The five-WP qualification workflow developed here provides a conceptual template for this extension, but the practical implementation raises non-trivial challenges. Walk-in chambers have test volumes being one to two orders of magnitude larger than the chambers studied here, making spatial uniformity characterisation with a chessboard grid of zinc tiles far more demanding. A minimum spatial resolution for the chessboard grid needs to be defined for this class of chambers, and the relationship between grid density and the ability to detect spatial non-uniformities needs to be validated. The calibration probe positioning requirements also need to be scaled appropriately.

Practical recommendations for zinc foil tile preparation

Two specific technical lessons learned during this study should be passed on explicitly to whoever repeats the zinc foil work. First, do not use hot moulding resin for cross-sectioning of the zinc tiles. Hot moulding requires temperatures in the range of 150–180 °C under pressure. The reason why zinc tile N14 was excluded is because it got partially deformed during the moulding process, which was stopped immediately after this result. Cold-setting epoxy resin is the only appropriate mounting medium for corroded zinc, and zinc-based specimens. Second, when applying the masking tape to the tiles, either ensure the tape seal is completely airtight around the perimeter, or deliberately leave a small drainage opening at the lowest corner of the tile during exposure. In some cases of this study, capillary action drew salt solution between the tape and the zinc surface on several tiles, creating an uncontrolled crevice geometry on the masked face that invalidated the mass loss measurements for those tiles.

Larger sample sizes and formal statistical inference

Throughout this thesis, formal significance testing (t-tests, ANOVA) was deliberately avoided because the sample sizes per subgroup were insufficient for reliable statistical inference. This was an explicit methodological decision, not an oversight, and the conclusions were instead based on the magnitude and consistency of observed differences. Future work should design the experimental programme from the outset with statistical power in mind.

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